Active Circuit Discovery: A Multi-Action POMDP Agent for Causal Feature Identification in Transformer Attribution Graphs

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Abstract: Mechanistic interpretability seeks to reverse-engineer the computational circuits within large language models, but current methods rely on exhaustive or heuristic search over exponentially many feature interactions. This paper introduces *Active Circuit Discovery* (ACD), a framework that combines attribution-graph analysis with active inference to select interventions efficiently. ACD uses Anthropic's `circuit-tracer` library as its attribution-graph backend, applying Edge Attribution Patching with transcoders to identify the active transcoder features for each prompt. A partially observable Markov decision process (POMDP) agent, implemented with `pymdp`, maintains a multi-factor generative model of feature importance, layer role, and causal influence. At each step the agent selects both a target feature and an intervention type (ablation, activation patching, or feature steering) by minimising Expected Free Energy over the joint feature–action space, and it learns its observation model online through Dirichlet parameter updates. ACD is an intervention-selection layer over existing attribution-graph tools; it is not a whole-circuit discovery method, and no claim of state-of-the-art circuit discovery is made. The framework is evaluated on Gemma-2-2B (26 layers) and Llama-3.2-1B (16 layers) across four settings: Indirect Object Identification (IOI), multi-step reasoning, feature steering, and a multi-domain benchmark spanning geography, mathematics, science, logic, and history. With a budget of 20 interventions per prompt, an ablation-only agent scored by bounded oracle efficiency against the ablation oracle reaches 82.0% efficiency on Gemma IOI and 73.0% on Gemma multi-step. It exceeds random selection by 43.5 percentage points on Gemma IOI (paired permutation $p = 0.031$) and is competitive with greedy ranking, a heuristic UCB bandit, and a plain UCB baseline. A direct Edge-Attribution-Patching ranking is itself a strong baseline that the agent does not consistently surpass, and on Llama multi-step the agent reaches 9.3% efficiency (37.8% with finer layer-role bins). All comparisons report bootstrap 95% confidence intervals. The full multi-action agent is characterised separately by a *Relative Cumulative KL*, a steering-driven amplification factor reported apart from the bounded efficiency. Feature steering changes the top-1 prediction in a dose-dependent manner, but a matched random-feature control shows that circuit-selected features are only marginally, and not significantly, more steerable than random active features at large multipliers, indicating that part of the effect is generic activation scaling. Multi-domain analysis shows task-dependent circuit structure, with IOI circuits concentrated in late layers and reasoning and scientific knowledge recruiting early and middle layers. Code, notebooks (free T4), amd64/aarch64 Docker images, and raw results are publicly available.

Keywords: mechanistic interpretability; active inference; circuit discovery; transcoders; Expected Free Energy; POMDP

1 Introduction

The rapid growth in the scale and deployment of large language models (LLMs) has intensified the need for principled methods to audit, explain, and predict their behaviour [1]. Mechanistic interpretability pursues this goal by reverse-engineering the internal computations of trained models into human-legible circuits [2, 3]. A circuit, in this context, denotes the minimal subgraph of model components (attention heads, MLP neurons, or SAE features) whose activations are jointly necessary and sufficient to reproduce a specific model behaviour on a given family of inputs [4, 5].

Circuit analysis has produced landmark findings. Wang et al. identified a 26-head circuit mediating Indirect Object Identification (IOI) in GPT-2 Small [5]; Hanna et al. localised a circuit for numerical greater-than comparisons [6]; and Nanda et al. traced the emergence of modular arithmetic generalisation through a Fourier-space circuit during grokking [7]. Each of these studies required thousands of manually guided interventions, prompting interest in automation [8, 9].

Automated Circuit Discovery (ACDC) [8] generalises the activation patching methodology of Wang et al. to a greedy graph-pruning algorithm that systematically tests every edge in the computational graph. Although ACDC achieves strong fidelity on the IOI task, the number of interventions scales as $O(E)$ in the number of graph edges, which for deep transformer models reaches the tens of thousands. Edge Attribution Patching (EAP) [9] reduces this cost by approximating patch effects with integrated gradients, yet it trades correctness for efficiency and may miss higher-order interactions. Neither method incorporates uncertainty about the circuit structure or adapts its search strategy based on what has already been discovered.

The novelty of this work lies in an uncertainty-guided intervention-selection layer, not in a new attribution backend. ACD is not a replacement for attribution-graph construction methods such as EAP or `circuit-tracer`; it consumes their output, a pruned attribution graph over transcoder features, and contributes a policy for choosing which feature to intervene on next, and how, under a fixed intervention budget. The novelty therefore lies in the adaptive search strategy rather than in how the candidate graph is built. To date, automated circuit discovery has pursued either exhaustive search (as in ACDC) or gradient-based ranking (as in EAPand Causal Tracing [10]), with Sparse Feature Circuits [11] and Dictionary Learning approaches [12] following similar gradient-plus-pruning paradigms. None of these methods explicitly maintains a belief distribution over circuit components or uses uncertainty to guide adaptive, budget-aware exploration. This poses a practical problem: in production systems the number of interventions is constrained by compute budgets, yet existing methods do not optimise for informativeness per intervention. The statistical literature on active learning and Bayesian experimental design [13] has long established that uncertainty-guided selection outperforms passive ranking when the goal is to resolve ambiguity with minimal queries. The present work brings this principle to circuit discovery: by formulating the problem as a Partially Observable Markov Decision Process (POMDP) and selecting interventions to minimise Expected Free Energy (EFE), the framework achieves information-theoretically principled adaptation that trades off exploration (resolving circuit uncertainty) with exploitation (confirming known important features).

Active Inference is a normative framework for perception and action in biological agents that unifies perception, learning, and planning under a single objective: the minimisation of variational free energy (equivalently, the maximisation of model evidence) [14, 15]. When extended to planning, the framework introduces the Expected Free Energy ($\mathcal{G}$), which decomposes into an epistemic term (expected information gain about hidden states) and a pragmatic term (expected alignment with preferences) [16, 17]. $\mathcal{G}$ minimisation naturally trades off exploration and exploitation: an agent selects actions that resolve uncertainty about the world while pursuing desired outcomes.

This work proposes that circuit discovery is precisely the kind of problem for which $\mathcal{G}$ minimisation is appropriate. The "world" is the causal graph of an LLM; the "actions" are ablation and activation-patching interventions; the "hidden states" are the identities and importances of circuit-critical features; and the "preference" is the rapid identification of a faithful, minimal circuit. An Active Inference agent that maintains a belief distribution over candidate circuit components and selects interventions to maximally resolve that uncertainty will, in theory, identify circuits with fewer total interventions than either random or gradient-ranked selection.

The contributions of this paper are as follows.

(C1) Active Inference formulation. Circuit discovery is cast as a partially observable Markov decision process (POMDP) with three hidden state factors (feature importance, layer role, causal influence), three observation modalities (KL divergence, activation magnitude, graph connectivity), and three actions (ablation, patching, steering). The agent is implemented using the `pymdp` library [18], with Expected Free Energy guiding intervention selection and Dirichlet parameter updates enabling online learning of the observation model.

(C2) Framework implementation. The Active Circuit Discovery (ACD) framework integrates Anthropic's `circuit-tracer` [19, 20] for Edge Attribution Patching with GemmaScope transcoders and the `feature_intervention` API for causally correct transcoder-level interventions.

(C3) Validated evaluation with action-matched and attribution baselines. The agent is compared against a heuristic bandit, greedy importance ranking, a direct Edge-Attribution-Patching (EAP) ranking baseline computed from `circuit-tracer` attribution scores, random selection, and an oracle upper bound, on two models (Gemma-2-2B and Llama-3.2-1B) across four benchmarks (IOI, multi-step reasoning, feature steering, and a five-domain cognitive benchmark). To isolate the contribution of selection from that of the intervention type, an ablation-only agent (POMDP-abl) is evaluated against the ablation oracle as the primary, fair comparison, and action-matched (steering-enabled) variants of the random, greedy, and bandit baselines are reported separately. A plain UCB baseline is also included to isolate the heuristic bandit's engineered inductive bias.

(C4) Causal controllability, with controls. Feature steering experiments show that scaling transcoder features changes predictions in a clear dose-dependent manner, far above chance. A random-feature control reveals that circuit-selected features are only marginally, and not significantly, more steerable than randomly chosen features at large multipliers, indicating that a substantial part of the effect is generic activation scaling rather than selection-specific causal control. The causal claim is therefore stated cautiously and the controls are reported in full.

(C5) Reproducibility artifacts. Three Google Colab notebooks (free T4 GPU), both an aarch64 (DGX Spark) and a standard `x86_64/amd64` Docker image, and raw experiment results (JSON) are released for full reproduction.

To the best of the authors' knowledge, this is the first work to cast circuit discovery as a POMDP with principled Expected Free Energy minimisation for uncertainty-guided adaptive intervention selection. Prior work has not combined uncertainty-aware feature importance estimation via belief-state evolution, a multi-action intervention space with learnable action-type selection, online generative-model learning via Dirichlet posterior updates, and transcoder-level causal interventions across the full feature space. These components together constitute a previously unexplored design point for the intervention-selection sub-problem. No claim is made of state-of-the-art circuit discovery relative to ACDC or EAP. On a shared attribution-graph backend and a fixed budget, EFE-guided selection (POMDP-abl) reliably beats random selection and is competitive with greedy, a heuristic UCB bandit, and a plain UCB baseline; a direct EAP attribution ranking is itself a strong ablation-KL baseline and is not consistently beaten (Section 5). The agent's distinctive value is therefore the explicit belief-state uncertainty, the multi-action space, and online model learning, capabilities the static baselines lack, rather than a raw efficiency win.

The following testable hypotheses guide the evaluation.

H1 (Efficiency). The POMDP agent, restricted to the ablation action for a fair comparison (the ablation-only variant "POMDP-abl"), identifies causally important features with higher cumulative KL divergence than random selection within a fixed budget of $B = 20$ interventions. This is assessed via a one-sided paired t-test and an exact one-sided paired permutation test across prompts ($\alpha = 0.05$), with the permutation test treated as primary given the small prompt counts.

H2 (Causal controllability). This hypothesis comprises two sub-claims. Under H2a (manipulability), scaling transcoder features changes the model's top-1 prediction far above chance, in a dose-dependent manner across the multiplier sweep (one-sided binomial test against a 1% chance baseline). Under H2b (selectivity), circuit-selected features are more controllable than matched randomly chosen features (one-sided Fisher exact test of selected versus control flip rates, $\alpha = 0.05$). Separating these sub-claims lets the controls reveal how much of the steering effect is specific to selected features versus generic activation scaling.

H3 (Discovery quality). The ablation-only agent (POMDP-abl) recovers $\geq 50\%$ of the ablation oracle's cumulative KL with budget $B = 20$ (bounded oracle efficiency). Assessed by point estimate with a bootstrap 95% confidence interval across prompts.

H4 (Cross-architecture transfer). The qualitative findings obtained on Gemma-2-2B replicate on Llama-3.2-1B.

The remainder of this paper is structured as follows. Section 2 reviews related work on circuit analysis, active learning and Bayesian experimental design, Active Inference, and sparse autoencoders. Section 3 describes the ACD framework in detail. Section 4 specifies the experimental protocol. Section 5 reports validated results. Section 6 provides an expanded discussion of findings, limitations, and future directions. Section 7 concludes. Additional implementation details and supplementary analyses are provided in the Appendix.

2 Background and Related Work

2.1 Mechanistic Interpretability and Circuit Analysis

Mechanistic interpretability traces causal pathways through neural networks by applying targeted interventions (zeroing activations, swapping activations between forward passes on different inputs, or replacing specific components with learned approximations) and measuring the resulting change in model output [21, 22]. The transformer architecture [23] is especially amenable to this analysis because its residual stream structure allows the contribution of each component to the final logits to be decomposed additively [3].

Olah et al. introduced the concept of circuits in convolutional vision networks [2, 4]. Elhage et al. formalised the residual stream decomposition for transformers and demonstrated in-context learning heads, induction heads, and name-mover heads in small two-layer models [3]. Wang et al. applied this framework at scale in a celebrated study of the 26-head IOI circuit in GPT-2 Small, using path patching to confirm causal roles [5]. Subsequent work characterised circuits for docstring completion [24], numerical reasoning [6], copy suppression in attention heads [25], and Chinchilla at 70 B parameters [26].

A recurring limitation of manual circuit analysis is labour intensity. Conmy et al. addressed this with Automated Circuit Discovery (ACDC), a greedy edge-pruning algorithm that recovers circuits matching manual analyses on the IOI and docstring tasks [8]. Syed et al. proposed Edge Attribution Patching (EAP), which approximates intervention effects with gradient-based attribution and achieves competitive results at a fraction of the runtime [9]. Geiger et al. introduced causal abstraction as a formal verification criterion for circuit hypotheses [22], and Chan et al. developed causal scrubbing for the same purpose [27].

Meng et al. complemented circuit analysis with causal tracing applied to factual associations in GPT models, identifying mid-layer MLP blocks as the primary locus of stored world knowledge [10]. Recent efforts have extended circuit analysis to sparse autoencoders and transcoders: Marks et al. combined gradient-based pruning with sparse features [11], while He et al. applied dictionary learning to discover interpretable circuits [12]. Lindsey et al. recently combined attribution graphs constructed via `circuit-tracer` [19] with large-scale SAE analysis to map the biology of Claude 3 Sonnet at a component level [28].

2.2 Sparse Autoencoders and Transcoders

Polysemanticity, the tendency of individual neurons to respond to multiple unrelated concepts, has been identified as a core obstacle to mechanistic interpretability [2, 29]. Sparse Autoencoders (SAEs) address this by learning an overcomplete, sparse linear basis for the residual stream at each layer. Cunningham et al. demonstrated that features learned by SAEs are substantially more monosemantic than individual neurons [30]. Bricken et al. scaled this approach to a one-layer MLP with 4,096-dimensional activations, recovering over 512 interpretable features including curve detectors and orientation-selective units [29]. Gao et al. studied scaling and evaluation of k-sparse SAEs at large scale, including on GPT-4 activations, and reported scaling laws for reconstruction and feature quality [31]. Rajamanoharan et al. proposed Gated Sparse Autoencoders, which improve feature monosemanticity by decoupling feature detection from magnitude estimation [32]. Paulo et al. demonstrated large-scale automated interpretation of millions of SAE features using large language models as interpretability oracles [33]. Makelov et al. provided principled evaluation criteria for SAE quality, showing that standard metrics can overestimate interpretability [34].

Transcoders are a related architecture that decompose MLP input-to-output transformations directly, providing a more causally faithful representation of MLP computation than residual-stream SAEs. Anthropic's `circuit-tracer` library [19, 20] provides an end-to-end pipeline for feature-level attribution: transcoders decompose MLP activations into sparse features, and Edge Attribution Patching constructs a directed graph of feature interactions. The `feature_intervention` API then allows causally correct ablation and steering of individual transcoder features with proper network propagation. GemmaScope [35, 36] provides pre-trained transcoders and SAEs for Google's Gemma-2 model family, enabling open interpretability research at scale.

2.3 Active Inference and Expected Free Energy

Active Inference is a unified computational framework derived from the Free Energy Principle [14], which posits that biological agents minimise the surprise (negative model log-evidence) associated with their sensory states. Friston et al. formalised this as variational inference: agents approximate the true posterior over hidden states $\boldsymbol{s}$ given observations $\boldsymbol{o}$ by maintaining a recognition distribution $q(\boldsymbol{s})$ and minimising the variational free energy $\mathcal{F} = D_{\mathrm{KL}}(q(\boldsymbol{s}) \,\|\, p(\boldsymbol{s} \mid \boldsymbol{o})) \geq -\log p(\boldsymbol{o})$ [37].

For planning and action selection, Da Costa et al. extended this to the Expected Free Energy, defined for a policy π over future time steps $\tau > t$ as [17]:

$$\mathcal{G}(\pi) = \sum_{\tau > t} \underbrace{\mathbb{E}\llbracket \left[\log p(\boldsymbol{o}_\tau) - \log q(\boldsymbol{o}_\tau|\pi)\right]}_{\text{pragmatic}} - \underbrace{\mathbb{E}\llbracket \left[\log q(\boldsymbol{s}_\tau|\pi) - \log q(\boldsymbol{s}_\tau|\boldsymbol{o}_\tau, \pi)\right]}_{\text{epistemic}} \tag{1}$$

The pragmatic value measures expected reward (preference satisfaction), while the epistemic value measures expected information gain (uncertainty reduction). Policies are selected according to a Boltzmann distribution over negative $\mathcal{G}$ values. This decomposition reveals how Active Inference naturally trades off exploration (maximising the epistemic term via information gain) and exploitation (optimising the pragmatic term via preference satisfaction).

As Parr et al. describe [15], Active Inference provides a unified model of cognition in which perception, learning, and action selection emerge from the same variational objective. Tschantz et al. demonstrated that $\mathcal{G}$ minimisation recovers reinforcement learning in the limit of pure pragmatic value [38].

The connection to Bayesian experimental design is particularly relevant for circuit discovery. MacKay [13] defined optimal experimental design as selecting experiments that maximise information gain about unknown parameters. The epistemic term in equation 1 is precisely this information gain (entropy reduction in the posterior belief). Thus, for problems where pragmatic preferences are absent or secondary, EFE minimisation reduces to maximising information gain—a principled, information-theoretic approach to adaptive experiment selection.

Heins et al. released `pymdp`, an open-source library that implements discrete POMDP agents with the full Active Inference formalism [18]. The implementation includes: (i) variational state inference using Bayesian belief updating, (ii) flexible observation and transition models specified as categorical and Dirichlet distributions, (iii) policy evaluation via EFE computation over a planning horizon, (iv) Dirichlet prior learning via parameter updates after each observation, and (v) action precision (inverse temperature) tuning for Boltzmann policy selection. The `pymdp` agent maintains a categorical posterior over discrete hidden state factors and updates it via Bayes rule after each observation. Critically, the observation model $p(\boldsymbol{o}|\boldsymbol{s})$ can be learned online: priors are stored as Dirichlet parameters, which are incremented after each observation to reflect new empirical counts. This allows the agent to refine its causal estimates iteratively. The present work uses `pymdp` to implement the circuit discovery agent with online learning of transcoder-level intervention effects.

Precision-weighted prediction-error dynamics are central to discrete-state active inference formulations [17, 37]. The present work operationalises this connection by embedding a pymdp POMDP agent within an automated circuit discovery procedure, leveraging EFE-guided action selection to adaptively discover circuits with minimal interventions.

2.4 Active Learning and Bayesian Experimental Design

The principle of selecting informative queries to resolve uncertainty is well- established in statistics and machine learning. MacKay [13] introduced information-theoretic criteria for Bayesian active learning, showing that agents should select experiments that maximise expected information gain (entropy reduction) about unknown parameters. This framework unifies optimal experimental design across domains: in Bayesian neural networks, active learning, and adaptive sampling.

Expected Free Energy minimisation generalises MacKay's information-theoretic principle to the full POMDP setting. The epistemic term in EFE (equation 1) is precisely the expected information gain about hidden states [16, 17]. When the pragmatic term is zeroed, EFE minimisation reduces to maximising information gain, that is, selecting the action that most reduces posterior uncertainty. This connection grounds Active Inference in the principled statistical tradition of Bayesian optimal design. The present work leverages this alignment: circuit discovery is cast as a Bayesian experimental design problem in which the "experiment" is a causal intervention (ablation, patching, or steering) and the "unknown parameters" are feature importances and roles. By minimising EFE, the discovery agent automatically balances exploration (resolving circuit uncertainty) with exploitation (confirming already-identified important features).

2.5 Gap Analysis

Table 1 synthesises the properties of existing automated circuit discovery methods and the proposed ACD framework. Current approaches fall into three categories: (1) *Exhaustive* methods (ACDC) test all possible interventions but lack any guidance from uncertainty estimates, scaling as $O(E)$ in graph size. (2) *Gradient-based* methods (EAP, Causal Tracing, Sparse Feature Circuits, Dictionary Learning) approximate intervention effects via attribution and rank candidates statically, with no online adaptation or uncertainty modeling. (3) *POMDP-based* methods (ACD) maintain a posterior belief over circuit components and adaptively select interventions to maximally reduce this uncertainty, grounded in information-theoretic principles. The novelty of ACD is threefold: it explicitly models uncertainty via a belief state distribution, uses that uncertainty to guide adaptive intervention selection via EFE, and learns the observation model online via Dirichlet parameter updates, enabling the agent to refine its causal estimates after each intervention. These categories are not mutually exclusive: ACD is complementary to ACDC and EAP rather than a competing graph-construction method. It consumes an attribution graph (here built with `circuit-tracer`'s EAP) and decides which of its features to test next under a budget. Accordingly, the fair point of comparison is not whole-circuit discovery but selection efficiency on a shared candidate set, measured against a direct EAP attribution ranking (Sections 4 to 5).

Table 1: Comparison of automated circuit discovery methods.

Method	Strategy	Uncert.	Adaptive	Gen. Mod.	Multi-act.	Online Learn.	Scale
ACDC [8]	Exhaustive	No	Threshold	No	No	No	GPT-2 Small
EAP [9]	Gradient	No	No	No	No	No	GPT-2 Small
Causal Tracing [10]	Gradient	No	No	No	No	No	GPT (various)
Sparse Features [11]	Gradient+Prune	No	No	No	No	No	GPT-2
Dict. Learning [12]	Gradient	No	No	No	No	No	Med. LMs
ACD (this work)	POMDP-EFE	Yes	Yes	Yes	Yes	Yes	2B+ params

3 The Active Circuit Discovery Framework

This section formalises the ACD framework, detailing the attribution graph backend, the POMDP-based active inference agent, and the intervention engine.

3.1 Architecture Overview

The ACD framework is organised as a three-layer pipeline in which an attribution backend supplies a pruned set of candidate transcoder features, an Active Inference POMDP agent chooses which feature to probe and which intervention to apply, and an intervention engine executes the chosen action and returns an observation that drives the next belief update. Figure 1 shows the end-to-end data flow between these three layers.

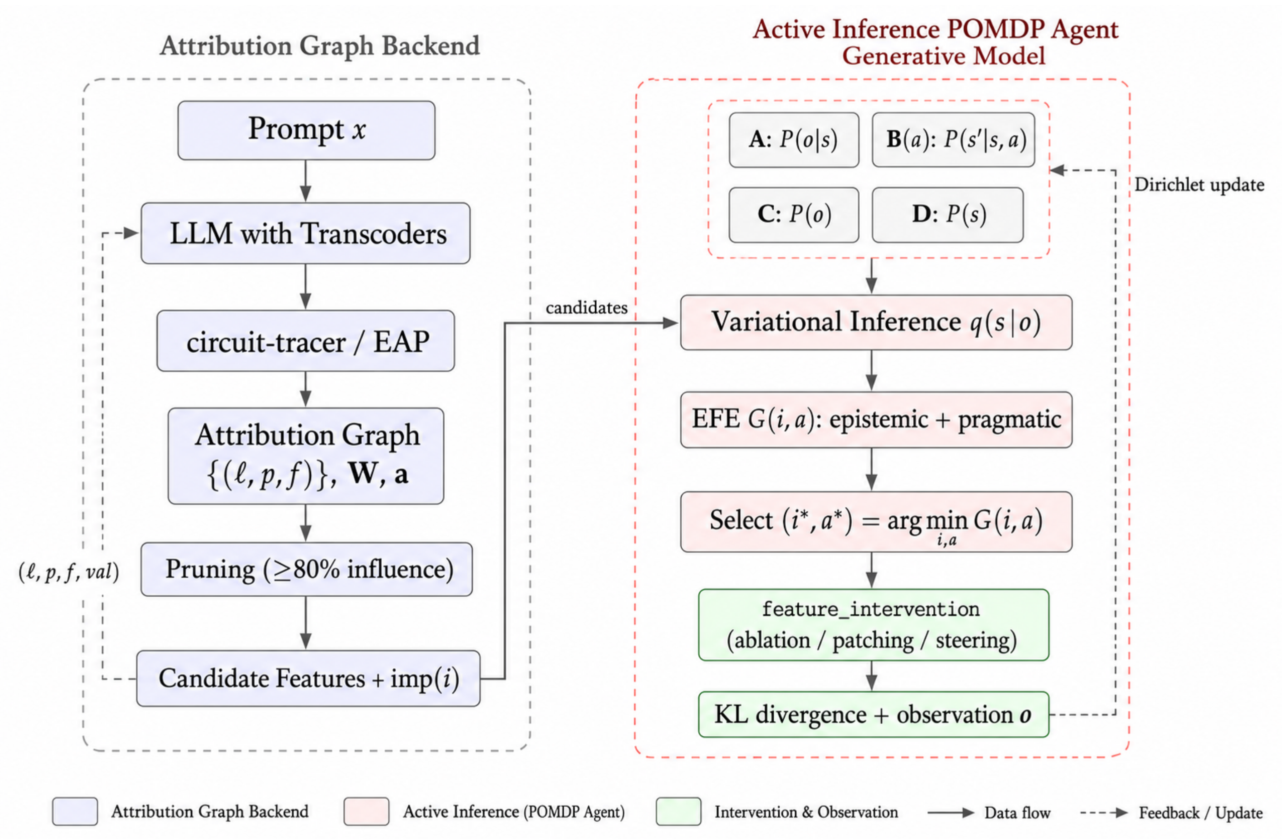

Figure 1: System architecture of ACD: attribution backend (left) and POMDP agent loop (right).

In more detail, the first layer is the *attribution graph backend*, in which Anthropic's `circuit-tracer` library [19] generates attribution graphs via Edge Attribution Patching (EAP) with GemmaScope transcoders for Gemma-2-2B; the graph contains active transcoder features, an adjacency matrix encoding feature interactions, and activation values. The second layer is the *Active Inference POMDP agent*, a multi-factor POMDP implemented with `pymdp` [18] that maintains a generative model of the circuit structure, performs variational state inference, evaluates candidate interventions via Expected Free Energy, and learns its observation model online from real intervention data. The third layer is the *intervention engine*, which executes feature-level ablations and steering via the `feature_intervention` API, intervening at the transcoder level with proper network propagation through the underlying TransformerLens [39] model.

3.2 Candidate Feature Extraction

Given a prompt, the attribution graph backend produces three outputs: active features $\{(l_i, p_i, f_i)\}$ where l is layer, p is token position, and f is feature index; an adjacency matrix $\mathbf{W} \in \mathbb{R}^{n\times n}$ encoding feature-to-feature attribution weights; and activation values $\boldsymbol{a} \in \mathbb{R}^n$.

The graph is pruned to retain features with influence above a threshold (default 80%). Each retained feature i receives a normalised importance score:

$$\mathrm{imp}(i) = \frac{\sum_j |W_{ij}| + \sum_j |W_{ji}|}{\max_k \left(\sum_j |W_{kj}| + \sum_j |W_{jk}|\right)} \tag{2}$$

Candidates are sampled across layers (up to k per layer) to ensure diversity.

3.3 POMDP Formulation

Circuit discovery is cast as a discrete Partially Observable Markov Decision Process (POMDP) in which each candidate feature carries a small set of unobserved attributes (importance, layer role, causal influence) and each intervention produces noisy, discretised observations that refine the agent's belief over those attributes. The overall inference–evaluation–action–learning loop is illustrated in Figure 2, and the individual factors are formalised in the subsections that follow.

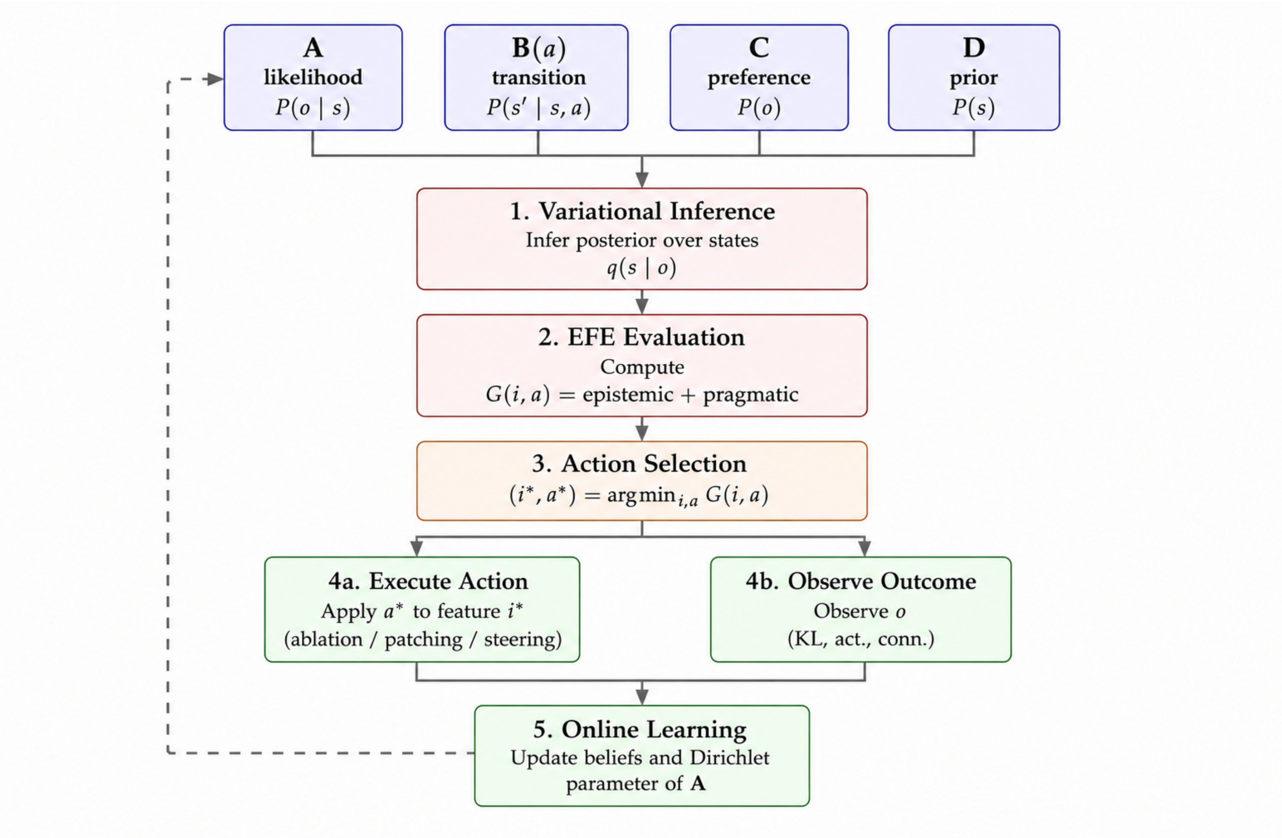

Figure 2: Active Inference POMDP agent pipeline: inference, EFE evaluation, and action selection.

3.3.1 Hidden State Factors

Three hidden state factors capture distinct aspects of each candidate feature. Factor 0 (*feature importance*, $s_0 \in \{\text{negligible}, \text{low}, \text{moderate}, \text{high}\}$, 4 levels) represents the causal contribution of a feature to the model's output on the current prompt. Factor 1 (*layer role*, $s_1 \in \{\text{early}, \text{middle}, \text{late}\}$, 3 levels) is determined by the feature's position within the network, with layers divided into thirds. Factor 2 (*causal influence*, $s_2 \in \{\text{weak}, \text{moderate}, \text{strong}\}$, 3 levels) reflects the feature's degree of causal control over downstream computation, inferred from intervention effects.

3.3.2 Observation Modalities

Three observation modalities provide complementary evidence about hidden states. Modality 0 (*KL divergence magnitude*) is the KL divergence between the clean and intervened output distributions, discretised into four levels: negligible ($< 10^{-4}$), small ($< 10^{-3}$), medium ($< 10^{-2}$), and large ($\geq 10^{-2}$). Modality 1 (*activation magnitude*) is the absolute activation value of the feature, also discretised into four levels. Modality 2 (*graph connectivity*) is the sum of in-degree and out-degree of the feature in the attribution graph, discretised into three levels (sparse, moderate, dense).

3.3.3 Actions

Three intervention types are available: *ablation* (action 0), which sets the transcoder feature activation to zero; *activation patching* (action 1), which replaces the feature activation with a mean-centred reference value; and *feature steering* (action 2), which scales the activation by a multiplier. Each action induces distinct transition dynamics in the B-matrix (see Section B): ablation is exploratory and shifts the importance belief downward, patching is confirmatory and surfaces real importance, and steering preserves the current belief. The agent selects which action to apply to each candidate by minimising Expected Free Energy over the joint (feature, action) space.

3.3.4 Generative Model

The generative model consists of four components, following the standard Active Inference formulation [15, 37]. The observation likelihood **A** specifies $P(o_m \mid s_0, s_1, s_2)$ for each modality m, encoding domain-informed priors about how hidden feature states produce observable intervention effects; this matrix is learned online via Dirichlet concentration updates from each observation. The transition model **B** specifies $P(s' \mid s, a)$ with action-conditioned dynamics: Factor 0 (importance) has distinct transition matrices for each of the three actions, encoding how ablation, patching, and steering affect the agent's belief about feature importance (see Appendix B for the calibrated priors). Factors 1 and 2 have identity transitions tiled across actions, reflecting intrinsic properties that do not change upon intervention. The preference model **C** is a log-prior over preferred observations in which higher KL divergence and higher activation are favoured, encoding the goal of finding causally important features. Finally, the prior **D** over hidden states is biased toward low importance, since most features are not causally critical, and uniform over layer role.

3.4 Intervention Selection via Expected Free Energy

In the single-step ACD setting, each policy π corresponds to a single candidate feature i and intervention type a; substituting $\pi = (i, a)$ into Equation 1 yields the following per-step EFE.

$$\mathcal{G}(i,a) = \underbrace{\mathbb{E}_q\big[\log q(s_\tau|\pi) - \log q(s_\tau|o_\tau,\pi)\big]}_{\text{epistemic}} + \underbrace{\mathbb{E}_q\big[\log p(o_\tau) - \log q(o_\tau|\pi)\big]}_{\text{pragmatic}} \quad (3)$$

The candidate–action pair with the lowest EFE is selected:

$$(i^*, a^*) = \underset{(i,a)}{\arg\min}\ \mathcal{G}(i,a) \quad (4)$$

This selection criterion naturally trades off exploration (choosing features that maximally reduce uncertainty about hidden states) with exploitation (choosing features that are expected to produce preferred observations, namely high KL divergence).

3.5 Belief Updating and Learning

After executing the selected intervention and observing the resulting KL divergence, activation magnitude, and graph connectivity, the agent performs three updates. First, it infers posterior states: the pymdp agent performs variational inference to update $q(s_0, s_1, s_2 \mid o_0, o_1, o_2)$. Second, it updates the observation model by incrementing the Dirichlet concentration parameters:

$$p_{A_m}(o_m, s_0, s_1, s_2) \mathrel{+}= \eta \cdot q(s_0) \cdot q(s_1) \cdot q(s_2) \quad (5)$$

where η is the learning rate; this enables the agent to improve its mapping from hidden states to observations as it accumulates intervention data. Third, it advances its internal time index and resets the observation buffer for the next step.

3.6 Convergence Detection

The agent monitors the KL divergence between successive belief distributions over a rolling window. When the average belief change falls below a threshold $\theta = 0.01$, the agent signals convergence, indicating that further interventions are unlikely to substantially revise the inferred circuit structure.

3.7 Intervention Engine

The intervention engine is the execution layer of ACD: it realises the abstract action selected by the POMDP agent as a concrete, causally correct manipulation of a single transcoder feature. At each step it (i) receives the chosen (feature, action) pair from the agent, (ii) dispatches it through the `feature_intervention` API at the exact (l, p, f) coordinate, (iii) measures the KL divergence between the clean and intervened next-token distributions, and (iv) returns the resulting observation tuple (KL magnitude, activation magnitude, graph connectivity) that drives the Dirichlet update of the observation model. The full per-step flow, including candidate extraction, EFE evaluation, intervention dispatch, and belief/parameter updates, is summarised in Figure 3.

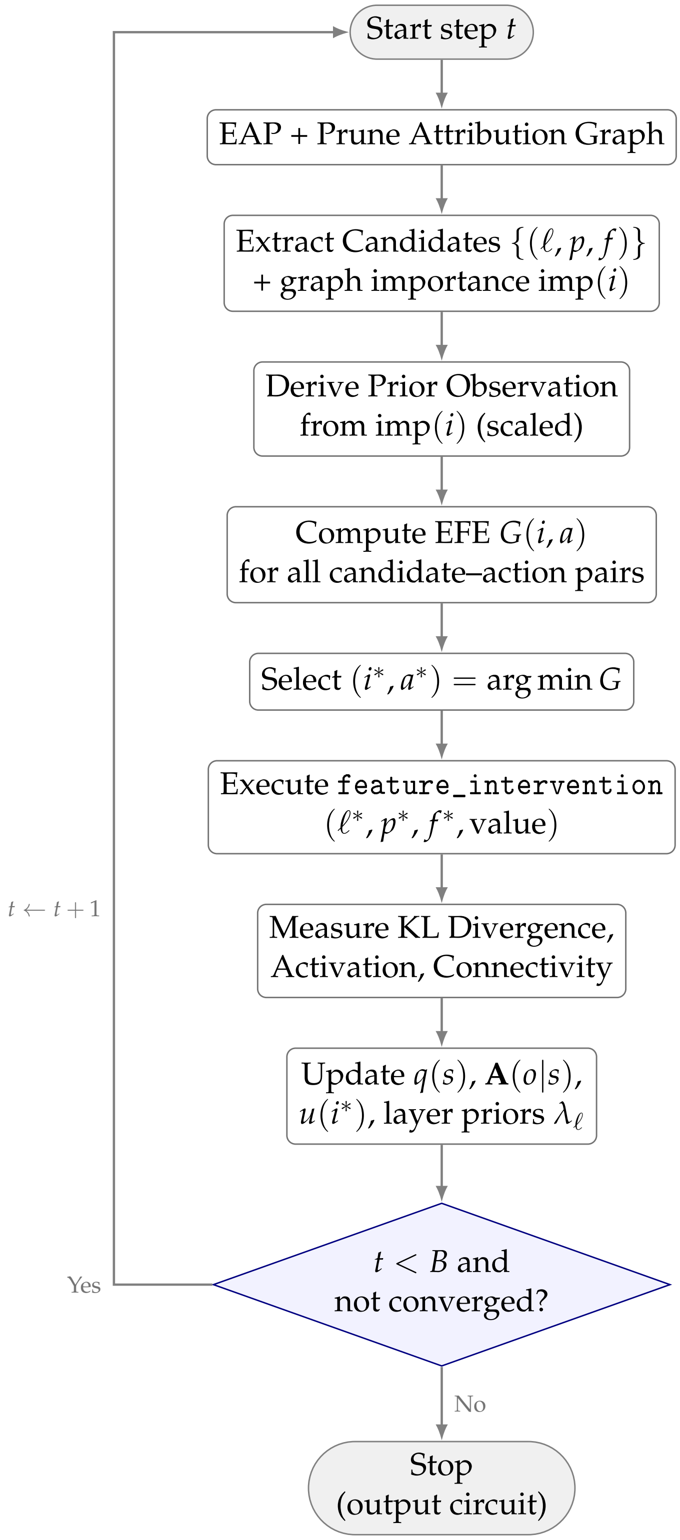

Figure 3: Flow of a single intervention step in Active Circuit Discovery.

The engine implements three manipulation types, each dispatched via the `feature_intervention` API according to the action selected by the POMDP agent at each step:

Ablation sets transcoder feature (l, p, f) to zero:

$$\text{logits}_{\text{abl}} = \text{feature\_intervention}(\text{prompt}, [(l, p, f, 0)]) \tag{6}$$

Activation patching replaces the feature activation with a mean-centred reference value r:

$$\text{logits}_{\text{patch}} = \text{feature\_intervention}(\text{prompt}, [(l, p, f, r)]) \tag{7}$$

Feature steering scales the activation by a multiplier m:

$$\text{logits}_{\text{steer}} = \text{feature\_intervention}(\text{prompt}, [(l, p, f, a_i \cdot m)]) \tag{8}$$

where a_i is the clean activation value of feature i.

Effect is measured via KL divergence between the clean and intervened output distributions:

$$\text{KL}_i = D_{\text{KL}}\big(\text{softmax}(\text{logits}_{\text{interv}}) \parallel \text{softmax}(\text{logits}_{\text{clean}})\big) \tag{9}$$

The `feature_intervention` API correctly propagates the intervention through the replacement model's transcoder structure, ensuring that the measured effect reflects the true causal contribution of the feature rather than a residual-stream approximation.

4 Experimental Setup

4.1 Models and Hardware

The framework is evaluated on two transformer architectures. Gemma-2-2B [35] has 26 layers, a 2304-dimensional hidden size, and 8 attention heads; its transcoders are drawn from GemmaScope (`mwhanna/gemma-scope-transcoders`). Llama-3.2-1B [40] has 16 layers, a 2048-dimensional hidden size, and 32 heads with grouped-query attention; its transcoders are from `mntss/transcoder-Llama-3.2-1B`. Both models use the `circuit-tracer` library [19] with the TransformerLens backend for attribution graph construction and transcoder-level interventions.

Experiments run on an NVIDIA DGX Spark with GB10 GPU (128 GB unified memory, CUDA 12.8, aarch64). Colab notebooks reproduce results on a free T4 GPU.

4.2 Benchmarks

4.2.1 IOI Circuit Recovery

The Indirect Object Identification task, introduced by Wang et al. [5], tests whether the agent can identify features causally responsible for predicting the indirect object. Five prompts of the form "When [A] and [B] went to the store, [A] gave the bag to __" are used, where the correct prediction is [B]. For each prompt, the KL divergence caused by ablating each candidate feature via the `feature_intervention` API is measured.

4.2.2 Multi-step Reasoning

The multi-step reasoning benchmark probes whether the discovery agent can localise features that mediate compositional inference rather than the single-hop associations that dominate IOI. Three prompts are used, each requiring the model to chain two or more intermediate facts before producing the target token: a transitive ordering prompt ("If Alice is taller than Bob, and Bob is taller than Carol, then the tallest person is"), a factual-chain prompt ("The capital of France is Paris. Paris is in Europe. The continent containing Paris is"), and a syllogistic prompt ("All dogs are animals. Fido is a dog. Therefore Fido is"). Unlike IOI, where causally critical features are concentrated in a small set of late-layer name-mover heads, compositional reasoning is expected to recruit earlier and middle layers that bind entities and propagate intermediate results through the residual stream. This shifts the effective search distribution away from the tail of the attribution graph and stresses the agent's ability to allocate its budget across layers rather than exploit a known late-layer prior. For each prompt the KL divergence between the clean and intervened next-token distributions is measured, using the same $B = 20$ budget and the same bandit, UCB, greedy, EAP, random, and oracle baselines as the IOI benchmark so that bounded oracle efficiency is directly comparable across tasks. This is also the benchmark used to evaluate the finer-layer-bin variant (Section 4.6).

4.2.3 Feature Steering

Five concept prompts (Golden Gate Bridge, Eiffel Tower, Mount Everest, Great Wall of China, Statue of Liberty). For each, the top 10 features by graph importance are identified and their activations scaled across a multiplier sweep $m \in \{0, 1.5, 2, 3, 5, 10\}$. Because large multipliers can push activations out of distribution, the protocol (a) includes smaller factors (1.5, 2, 3) to characterise the dose-response curve, and (b) adds a random-feature control: a matched set of randomly chosen lower-importance features steered at the same multipliers. Causal control is evidenced only to the extent that circuit-selected features shift predictions more than this control. For every steered feature the top-five next-token probabilities are recorded at each multiplier, so that concept amplification at large factors can be distinguished from generic distributional collapse. The KL divergence and whether the model's top-1 prediction changes are measured for both sets.

4.2.4 Multi-Domain Benchmark

The framework is also evaluated across five cognitive domains with two prompts each: geography (e.g. "The capital of France is"), mathematics (e.g. "The square root of 64 is"), science (e.g. "Water is made of hydrogen and"), logic (syllogistic reasoning), and history (e.g. "The year World War II ended was"). This allows cross-domain comparison of circuit structure and layer distribution.

4.3 Experimental Prompts

All prompts used in the experiments are listed below. These are defined in the experiment runner script (`run_real_experiments.py`) and correspond exactly to the prompts whose results appear in the JSON outputs under `results/`.

The five IOI prompts are:

> "When John and Mary went to the store, John gave the bag to" "After Alice and Bob finished lunch, Alice handed the receipt to" "While Sarah and Tom were at the park, Sarah threw the ball to" "When Emma and David arrived at the office, Emma passed the keys to" "As Lisa and Mike left the restaurant, Lisa returned the coat to"

The three multi-step reasoning prompts are:

> "If Alice is taller than Bob, and Bob is taller than Carol, then the tallest person is" "The capital of France is Paris. Paris is in Europe. The continent containing Paris is" "All dogs are animals. Fido is a dog. Therefore Fido is"

The five feature-steering concept prompts are:

"The Golden Gate Bridge is" "The Eiffel Tower is located in" "Mount Everest is the tallest" "The Great Wall of China was built" "The Statue of Liberty stands in"

The ten multi-domain prompts, two per domain, are grouped by domain below. Geography prompts:

"The capital of France is" "The Golden Gate Bridge connects San Francisco to"

Mathematics prompts:

"The square root of 64 is" "If 2 + 3 = 5 then 3 + 4 ="

Science prompts:

"Water is made of hydrogen and" "The speed of light is approximately"

Logic prompts:

"All mammals are warm-blooded. A whale is a mammal. Therefore a whale is" "All birds have wings. A penguin is a bird. Therefore a penguin has"

History prompts:

"The year World War II ended was" "The first person to walk on the moon was"

4.4 Baselines

Baselines are grouped into ablation-only selectors (which, like POMDP-abl, only ablate and are therefore directly comparable to the ablation oracle) and action-matched selectors (which, like the full POMDP, may also steer, and are used only for the secondary RCK analysis of Section 5).

Among the ablation-only baselines, *Random* selection draws features uniformly at random (Equation 10; $N = 10$ trials, averaged):

$$a_t \sim \text{Uniform}(\mathcal{F}) \tag{10}$$

Greedy selection ranks features in descending order of graph node influence (sum of absolute adjacency weights), per Equation 11:

$$i^* = \arg\max_i \text{imp}(i) \tag{11}$$

where $\text{imp}(i)$ is the sum of absolute adjacency weights for node i (defined in Equation 2). *EAP* ranking is a direct Edge-Attribution-Patching baseline: candidates are ordered by the magnitude of their *direct attribution to the logit nodes* of the `circuit-tracer` graph, i.e. the feature's column summed over the logit rows of the adjacency matrix, $\text{eap}(i) = \sum_{t \in \mathcal{L}} |W_{t,i}|$ (normalised to $[0, 1]$). This isolates the classic EAP "direct effect on the output" signal, distinct from the total bidirectional connectivity used by Greedy, and provides an established-method point of comparison. The *bandit* selector combines graph importance with a decaying uncertainty bonus and learned per-layer priors, providing a comparison point that separates the contribution of Active Inference from simpler uncertainty heuristics:

$$S(i) = \text{imp}(i) \cdot \lambda_\ell + \omega_e \cdot u(i) \tag{12}$$

where $u(i)$ is the uncertainty bonus (initialised to 1, decayed on visit) and λ_ℓ is the per-layer prior updated from observations. A *plain UCB* baseline removes these engineered priors and instead applies the textbook UCB1 rule over layers, using the observed KL as reward; it isolates how much of the heuristic bandit's performance comes from its hand-built importance and locality priors rather than from upper-confidence exploration alone. The *oracle* selects features in descending order of true KL divergence, an upper bound that requires all ablations in advance:

$$i^* = \arg\max_i \mathrm{KL}_i^{\mathrm{abl}} \tag{13}$$

The action-matched baselines separate the contribution of the agent's selection policy from that of the steering action. Steering-enabled counterparts of the heuristics are evaluated that use the identical selection order but apply the feature-steering intervention: *Greedy+steer*, *Bandit+steer*, and *Random+steer*. A *Random-action* baseline that selects a feature uniformly at random and chooses its intervention type uniformly at random is also included, isolating the value of learned action selection. (Activation patching to the zero reference coincides with ablation in this setup, so this control effectively samples between ablation and steering.) The corresponding *multi-action oracle* ranks features by $\max(\mathrm{KL}_i^{\mathrm{abl}}, \mathrm{KL}_i^{\mathrm{steer}})$.

All methods use the same `feature_intervention` API and KL divergence metric. Budgets are fixed at $B = 20$ interventions per prompt.

4.5 Metrics

Metrics of two kinds are reported, and they are kept strictly separate because they live on different scales: a bounded efficiency for ablation-only selectors and a relative amplification factor for steering-enabled selectors.

Mean KL per intervention is the average KL divergence across selected features; higher values indicate that the method finds more causally important features. *Cumulative KL* (CumKL) is the total information gained over the budget.

(i) Bounded oracle efficiency. For *ablation-only* methods (POMDP-abl, Random, Greedy, EAP, Bandit, UCB), every selector and the oracle draw from the same quantity, ablation KL, so their ratio to the ablation oracle is bounded in $[0, 100]\%$ and is interpretable as an efficiency:

$$\eta = \frac{\mathrm{CumKL}_{\mathrm{agent}}^{\mathrm{abl}}}{\mathrm{CumKL}_{\mathrm{oracle}}^{\mathrm{abl}}} \times 100\%, \qquad \eta \in [0, 100]\%. \tag{14}$$

This is the primary discovery-quality metric.

(ii) Relative Cumulative KL (RCK). The full multi-action agent may steer, and steering can produce larger KL shifts than ablation; comparing its cumulative KL to the *ablation* oracle is therefore a ratio of two different quantities, not a bounded efficiency, and can exceed 100%. This quantity is named the Relative Cumulative KL,

$$\mathrm{RCK} = \frac{\mathrm{CumKL}_{\mathrm{agent}}}{\mathrm{CumKL}_{\mathrm{oracle}}^{\mathrm{abl}}} \times 100\%, \tag{15}$$

and report it only for the action-matched analysis, explicitly labelled as a relative amplification factor rather than an efficiency. All action-matched baselines (Greedy+steer, Bandit+steer, Random+steer, Random-action) are reported on the same RCK scale so the comparison is fair. The KL divergence per intervention is computed as in Equation 16:

$$D_{\mathrm{KL}}(p \,\|\, q) = \sum_t p(t) \log \frac{p(t)}{q(t)} \tag{16}$$

The *prediction change rate* is the fraction of steering experiments in which the model's top-1 prediction changes.

In summary, the primary discovery-quality claim uses the bounded oracle efficiency of Equation 14 applied to the ablation-only agent (POMDP-abl) against the ablation oracle, so that all quantities are on the same scale and the metric cannot exceed 100%. The RCK of Equation 15 is reported separately as a relative amplification factor for the full multi-action agent and its action-matched baselines, never as an efficiency. The Correspondence Metric is formally defined in Appendix D.

4.6 Statistical Analysis and Sensitivity

Given the modest prompt counts and the high coefficient of variation of per-prompt KL, every reported mean, bounded efficiency, and RCK is accompanied by a percentile bootstrap 95% confidence interval obtained by resampling prompts with replacement (10^4 resamples). Hypothesis H1 is assessed by both a one-sided paired t-test and an exact paired-permutation test of agent-versus-random per-prompt mean KL; a result is reported as significant only when $p < 0.05$ under the permutation test. H1 is not accepted where it is not significant (for example, Llama IOI).

The discretisation sensitivity of the agent is assessed at $\pm 20\%$. The agent discretises continuous observations (KL, activation, connectivity) at fixed thresholds, and maps graph importance into the KL-prior range through a 0.01 scaling factor (Appendix C). To show that conclusions are not artefacts of these choices, the Gemma IOI benchmark is re-run with the KL and activation thresholds independently scaled by 0.8 and 1.2, and the resulting change in bounded efficiency is reported. The scaling is exposed through the experiment runner flags `--kl-threshold-scale`, `--act-threshold-scale`, and `--imp-scale`.

To characterise robustness to the agent's core hyperparameters, the Gemma and Llama IOI benchmarks are additionally re-run while independently varying the action precision ($\gamma \in \{4, 16, 64\}$), the Dirichlet likelihood learning rate ($\eta \in \{0.5, 1.0, 2.0\}$), and the pragmatic preference weight ($\{0.5, 1.0, 2.0\}$), one factor at a time about the default. These factors are exposed through `--gamma`, `--lr-pa`, and `--pragmatic-weight`, and the resulting bounded efficiencies are reported in Section 5.

To probe the Llama multi-step result, a variant is additionally run in which the agent's single layer-role factor is refined from 3 bins (early/middle/late) to 6, via `--n-layer-role 6`, so that deeper layer structure can be represented in the belief state. The outcome is reported in Section 5 regardless of whether it improves the result.

4.7 POMDP Agent Configuration

The Active Inference POMDP agent uses 4 importance levels $\times$ 3 layer roles $\times$ 3 causal influence levels for a total of 36 joint hidden states. Observations comprise 4 KL levels, 4 activation levels, and 3 connectivity levels. The agent selects among 3 actions (ablation, patching, steering) using a single-step lookahead policy with stochastic selection via softmax over negative EFE ($\gamma = 16.0$). The observation model is learned online through Dirichlet updates on all modalities ($\eta = 1.0$), and inference uses the vanilla variational scheme implemented in pymdp.

4.8 Reproducibility

All code is available at `https://github.com/SharathSPhD/ActiveCIrcuitDiscovery`. Colab notebooks reproduce the main experiments on a free T4 GPU. The experiment runner supports model selection via `--model gemma|llama|both` and benchmark selection via `--experiment ioi|steering|multistep|domain|all`. Raw experiment outputs are stored in `results/` as JSON.

The ACD framework ships with two Docker targets so that results are not tied to a single architecture. `Dockerfile.dgx-spark` (with `docker-compose.yml`) targets the NVIDIA DGX Spark platform on which the headline numbers were produced (GB10 GPU, 128 GB unified memory, CUDA 12.8, aarch64). `Dockerfile.amd64` (with `docker-compose.amd64.yml`) provides a standard `x86_64` image built on the upstream `pytorch/pytorch:2.4.0-cuda12.1` base; it runs on any NVIDIA GPU of compute capability $\geq$ 7.5, including the 16 GB T4 offered for free on Google Colab. Because the model loads in float32 and uses only $\sim$5 GB of VRAM, the full pipeline fits on a single T4, and the accompanying Colab notebooks reproduce the IOI, steering, multi-step, and domain experiments end-to-end on that hardware. Both images install `circuit-tracer` from source and pin all remaining dependencies via `requirements.txt`; absolute KL values can shift marginally across GPU generations and driver/cuDNN versions, but the relative orderings and statistical conclusions reported here are hardware-independent.

Gemma-2-2B is available from HuggingFace Hub under the Gemma license (`google/gemma-2-2b`), and GemmaScope transcoders are fetched automatically by the `circuit-tracer` library. Experiments require a single NVIDIA GPU with at least 8 GB VRAM (tested on T4 with 16 GB in Colab and GB10 with 128 GB on DGX Spark). The model loads in float32 and requires approximately 5 GB VRAM. All dependencies are pinned in `requirements.txt`, with key packages including `circuit-tracer` (from git), `transformer-lens`, and `torch>=2.0`.

Random baselines use `numpy.random.seed(42)`, while the Active Inference Selector is deterministic given the same candidates and exploration weight. Approximate runtime is 40–60 seconds per prompt on a single GPU (attribution graph generation: $\sim$18s; 40 ablations at $\sim$30ms each: $\sim$1.2s; overhead: $\sim$20s for model setup on first prompt). All experiment outputs are saved as JSON in the `results/` directory, including per-prompt KL values for all methods, feature IDs, layer distributions, and steering outcomes.

5 Results

5.1 IOI Circuit Recovery

Intervention-selection quality is first evaluated on the Indirect Object Identification (IOI) task: 5 prompts with a budget of $B = 20$ feature-level interventions per prompt. The fair comparison fixes the intervention type to ablation for every method, so that all strategies are scored on the same ablation-KL scale against an ablation oracle that greedily ranks the candidate set by true ablation KL (Section 4). Under this protocol the agent is the ablation-constrained variant *POMDP-abl*, which uses the full Expected Free Energy (EFE) objective to choose which feature to test but is denied the steering action. The multi-action agent that also chooses the intervention type is analysed separately in Section 5.2. Mean and median per-step KL and the bounded oracle efficiency (cumulative method KL divided by cumulative oracle KL, $\leq$ 100% by construction) are reported with bootstrap 95% confidence intervals over prompts (Table 2).

Table 2: IOI feature discovery under the fair ablation-only protocol: per-prompt mean KL, median, inter-quartile range [Q1, Q3], and bounded oracle efficiency with bootstrap 95% CIs over 5 prompts.

Model	Method	Mean KL	Median	[Q1, Q3]	Oracle Eff. [95% CI]
Gemma-2-2B	Oracle	0.000862	—	—	100.0%
	EAP	0.000793	0.000436	[0.000399, 0.001306]	91.9% [82.1, 96.5]
	POMDP-abl	0.000707	0.000404	[0.000335, 0.001115]	82.0% [73.8, 87.8]
	Bandit	0.000642	0.000408	[0.000294, 0.001145]	74.4% [67.9, 79.0]
	UCB	0.000635	0.000394	[0.000242, 0.001155]	73.7% [59.0, 82.2]
	Greedy	0.000577	0.000492	[0.000260, 0.000583]	66.9% [47.5, 83.0]
	Random	0.000493	0.000289	[0.000202, 0.000867]	57.1% [49.0, 60.2]
Llama-3.2-1B	Oracle	0.009755	—	—	100.0%
	EAP	0.009291	0.005739	[0.003297, 0.018357]	95.2% [87.0, 97.7]
	Bandit	0.008606	0.005999	[0.003302, 0.014895]	88.2% [79.0, 96.7]
	UCB	0.005389	0.001458	[0.001227, 0.006000]	55.2% [10.5, 92.7]
	POMDP-abl	0.005086	0.002679	[0.001923, 0.006294]	52.1% [17.4, 90.2]
	Random	0.004720	0.001795	[0.001504, 0.007315]	48.4% [31.6, 64.7]
	Greedy	0.003982	0.001923	[0.000899, 0.002689]	40.8% [11.7, 74.1]

The efficiency denominator is the oracle's per-step mean KL, 0.000862 (Gemma) and 0.009755 (Llama), equivalently a cumulative KL of 0.0172 (Gemma) and 0.1951 (Llama) over the 20-step budget. On Gemma, POMDP-abl reaches 82.0% of the ablation oracle, clearly above random (57.1%) and greedy (66.9%) and modestly above the heuristic bandit (74.4%) and the plain UCB baseline (73.7%). A one-sided paired permutation test confirms the gain over random (+43.5% relative, $p = 0.031$, $n = 5$). A direct EAP attribution ranking is itself a strong ablation-KL baseline (91.9%) and is not surpassed by the agent. On Llama the picture is less favourable: POMDP-abl (52.1%) only matches random (48.4%) and the plain UCB baseline (55.2%), and trails both the heuristic bandit (88.2%) and EAP (95.2%); the gain over random is small and not significant (+7.8%, $p = 0.41$). No efficiency win is therefore claimed over the strongest ablation baselines, since EAP ranking is competitive or better in every tested setting; the agent's contribution is instead positioned as an adaptive, uncertainty-aware intervention-selection policy (Sections 5.2 and 6).

The temporal profile under the fair protocol is shown in Figure 4. The cumulative ablation-KL curves of all ablation-only methods are plotted against the oracle envelope: on Gemma, EAP tracks the oracle most closely, with POMDP-abl, bandit and greedy bunched below it; on Llama the bandit and EAP lead while POMDP-abl tracks greedy and random. No method exceeds the oracle, as expected when all methods share the ablation action.

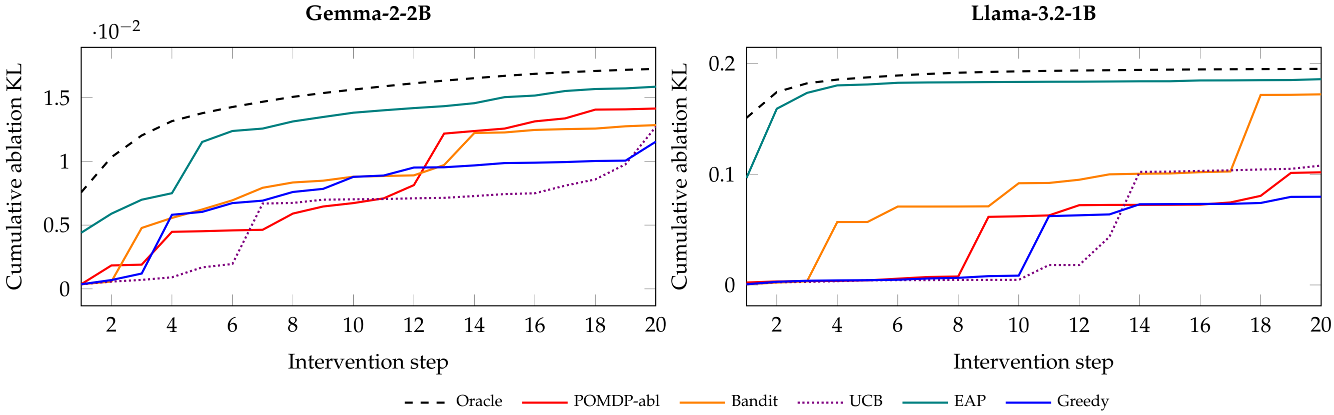

Figure 4: Cumulative ablation KL over 20 IOI intervention steps (Gemma left, Llama right). All curves are ablation-only and bounded above by the oracle (dashed).

Across all prompts, the top causal features are located in layers 24–25 (e.g., `L25_P14_F4717`, KL=0.0015–0.013), consistent with late-layer name-mover circuits identified in prior work by Wang et al. [5]. Figure 5 visualises representative attribution subgraphs around these features for both models and shows that the late-layer name-mover motif is preserved across the two architectures despite their different depths and hidden dimensions.

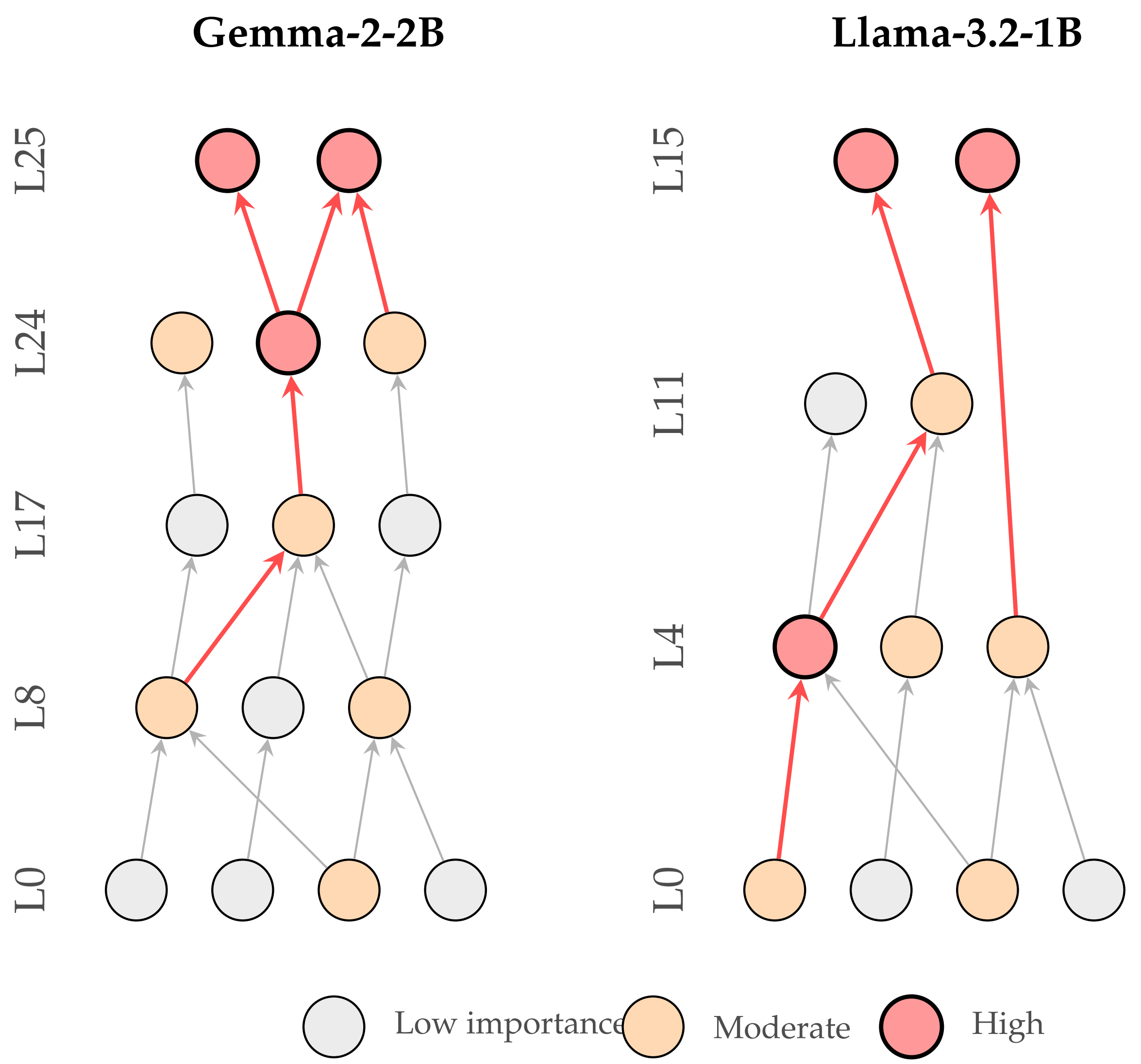

Figure 5: Simplified attribution graphs for IOI: Gemma-2-2B (left) and Llama-3.2-1B (right).

Figure 6 summarises bounded oracle efficiency for the five ablation-only methods across IOI and multi-step tasks. EAP is the strongest baseline on both models; POMDP-abl is competitive with the bandit on Gemma but falls behind on Llama, especially for multi-step reasoning (discussed in Section 5.5).

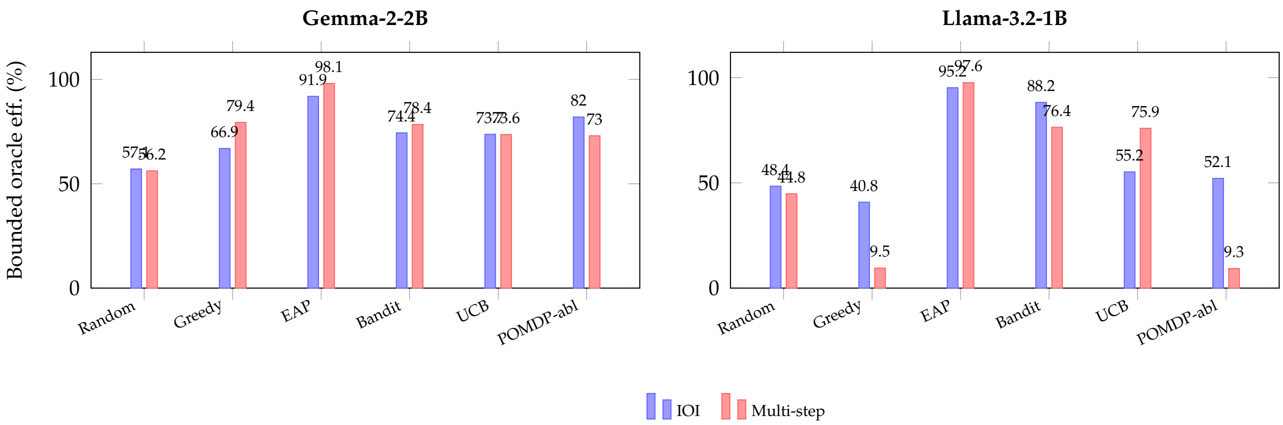

Figure 6: Bounded oracle efficiency (ablation-only) across IOI and multi-step tasks for Gemma-2-2B (left) and Llama-3.2-1B (right). By construction no method exceeds 100%.

5.2 Multi-Action Amplification: Relative Cumulative KL

When the agent is allowed to choose the intervention *type*, it overwhelmingly selects feature steering, which scales a feature's activation rather than zeroing it and can therefore produce KL divergences far larger than ablation on the same feature. Comparing this multi-action cumulative KL to the ablation oracle yields ratios well above 100%, so it is not an oracle efficiency. It is reported under a distinct name, the Relative Cumulative KL (RCK): the ratio of a multi-action method's cumulative KL to the cumulative KL of the ablation oracle (Section 4). To isolate how much of this amplification comes from selection rather than from simply having the steering action, the full POMDP is compared against action-matched baselines that share the identical multi-action space (greedy+steer, bandit+steer, random+steer) and a pure random-action control (Table 3).

Table 3: Relative Cumulative KL (RCK, %) of the multi-action POMDP versus action-matched baselines, relative to the ablation oracle (100%), with bootstrap 95% CIs over prompts.

Model	Method	IOI RCK [95% CI]	Multi-step RCK [95% CI]
Gemma-2-2B	POMDP (multi)	1255 [974, 1671]	979 [792, 1374]
	Bandit+steer	1185 [780, 1797]	1209 [798, 1935]
	Random+steer	888 [672, 1182]	737 [496, 1073]
	Greedy+steer	881 [605, 1675]	1223 [707, 1907]
	Random-action	371 [264, 528]	249 [213, 296]
Llama-3.2-1B	Bandit+steer	1467 [329, 2848]	279 [203, 412]
	Random+steer	961 [181, 1907]	381 [115, 868]
	Random-action	429 [86, 811]	164 [68, 340]
	Greedy+steer	353 [244, 482]	140 [55, 293]
	POMDP (multi)	91 [27, 320]	34 [19, 56]

The decomposition is revealing. On Gemma, every steering-enabled strategy, including random+steer (888%), produces RCK far above 100%, so the bulk of the amplification is attributable to the steering action itself, not to which feature is selected. The full POMDP (1255%) does sit at the top of the IOI ordering, indicating that EFE selection adds a further margin over random steering, though the action choice dominates. On Llama the agent behaves very differently: it mostly chooses ablation (70/100 actions, Section 5.4) and its RCK is only 91%, below the ablation oracle and well below bandit+steer (1467%). This confirms that the figures above 100% are a property of the steering intervention rather than evidence that the agent recovers more circuit structure than ablation-based methods; the RCK scale must therefore not be read as a super-oracle discovery rate. Figure 7 visualises the decomposition.

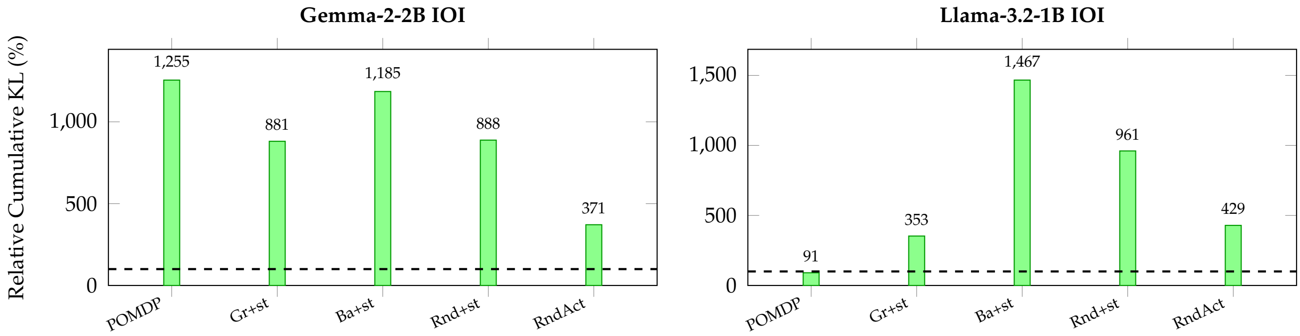

Figure 7: Relative Cumulative KL of the multi-action POMDP against action-matched baselines (Gr+st, Ba+st, Rnd+st) and a random-action control (RndAct), relative to the ablation oracle (100%, dashed).

5.3 Feature Steering and Causal Controllability

Causal controllability is tested by scaling individual transcoder feature activations across a multiplier sweep $m \in \{0, 1.5, 2, 3, 5, 10\}$ on 5 concept prompts with 10 circuit-selected features each (50 features per model). To separate selectivity from out-of-distribution (OOD) effects of large multipliers, a matched random-feature control of 10 random active features per prompt is subjected to the identical sweep. At each multiplier, the number of top-token prediction changes and the mean KL divergence are recorded for selected and control features (Table 4).

Table 4: Feature steering dose–response: prediction changes (selected / control, out of 50) and mean KL across the multiplier sweep. Circuit-selected features vs. a random-feature control.

Model	**Quantity**	$m{=}0$	$m{=}1.5$	$m{=}2$	$m{=}3$	$m{=}5$	$m{=}10$
Gemma-2-2B	Changes (sel.)	0/50	0/50	1/50	2/50	6/50	11/50
	Changes (ctrl.)	0/50	0/50	1/50	1/50	2/50	8/50
	Mean KL (sel.)	0.001	0.000	0.002	0.007	0.039	0.367
	Mean KL (ctrl.)	0.001	0.000	0.001	0.005	0.020	0.086
Llama-3.2-1B	Changes (sel.)	0/50	0/50	0/50	1/50	1/50	9/50
	Changes (ctrl.)	0/50	0/50	0/50	0/50	0/50	6/50
	Mean KL (sel.)	0.005	0.001	0.003	0.011	0.043	0.232
	Mean KL (ctrl.)	0.001	0.000	0.000	0.002	0.010	0.120

Two effects are visible. First, manipulability (H2a): both KL and prediction-change rate rise monotonically with the multiplier, and at $m{=}10$ the selected-feature change rate (11/50 on Gemma, 9/50 on Llama) is far above the per-token chance rate (a one-sided binomial test against chance gives $p < 10^{-8}$). Steering transcoder features causally and dose-dependently moves the output distribution. Second, and more cautiously, selectivity (H2b): circuit-selected features carry higher mean KL than random controls at large multipliers (for example, 0.367 versus 0.086 on Gemma at m=10), but the prediction-change advantage is small (11 versus 8 on Gemma, 9 versus 6 on Llama). A one-sided Fisher exact test on the selected-versus-control change counts is not significant, either at m=10 or pooled over $m \geq 2$. Circuit-selected features are thus only marginally, and not significantly, more controllable than random active features at the prediction level, and the largest multipliers push activations off-distribution; the steering result should accordingly be read as a controllability demonstration, not as proof of privileged selectivity.

A complementary view of the largest multipliers confirms that steering amplifies coherent concepts rather than merely degrading the model. As the multiplier increases from 0 to 10, the mean probability mass on the steered top-1 token remains high (approximately 0.43 on Gemma). At the same time, the set of distinct dominant tokens broadens from 5 to 11 and increasingly features content tokens associated with the steered concepts (for example geographic and landmark tokens) rather than collapsing onto generic function words. This indicates that large multipliers redirect the prediction toward specific concepts rather than producing undifferentiated noise.

The per-multiplier KL response is visualised in Figure 8.

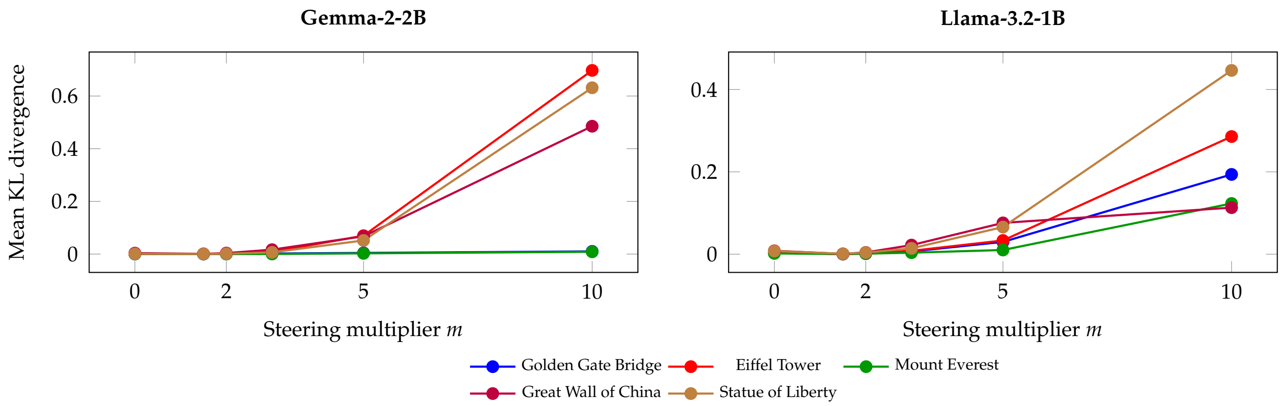

Figure 8: Mean KL divergence across steering multipliers for five concept prompts on both models.

5.4 Active Discovery Dynamics

The POMDP agent maintains explicit beliefs over three hidden state factors and updates them after each intervention. Across IOI prompts, total belief entropy decreases over the budget, indicating information accumulation about circuit structure. The online learning of the observation model itself also converges: the mean per-step L_1 drift of the learned KL-likelihood matrix, $\|A_t - A_{t-1}\|_1$, decreases monotonically over the budget (Figure 9), roughly halving between the first and last update on Gemma IOI (from 1.03×10^{-3} to 5.0×10^{-4}). The Dirichlet posterior over the observation model thus stabilises as evidence accumulates, rather than oscillating.

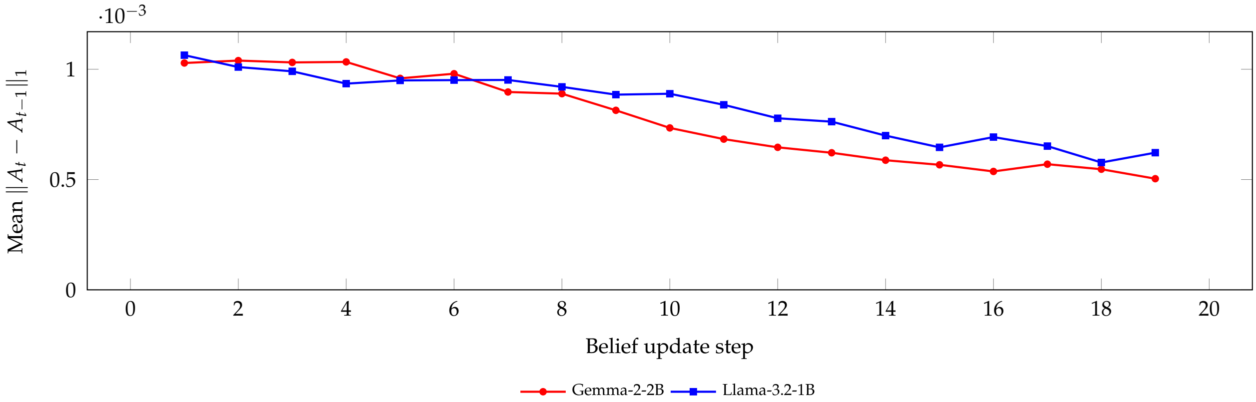

Figure 9: Online convergence of the learned observation model: mean per-step L_1 drift of the KL-likelihood matrix across IOI prompts.

The agent's action selection differs sharply between architectures, which explains the RCK contrast in Section 5.2. On Gemma IOI, ablation is selected first for exploration, after which the agent transitions to predominantly feature steering (94/100 steps; 94/95 post-first-step) with occasional activation patching (1/100). On Llama IOI the agent is far more conservative, choosing ablation for 70/100 steps and steering only 28/100, so its cumulative KL stays near the ablation scale rather than amplifying. The step-by-step action distribution for both models is shown in Figure 10. This action-policy divergence, rather than any difference in feature ranking, is what drives the very different RCK values across the two models.

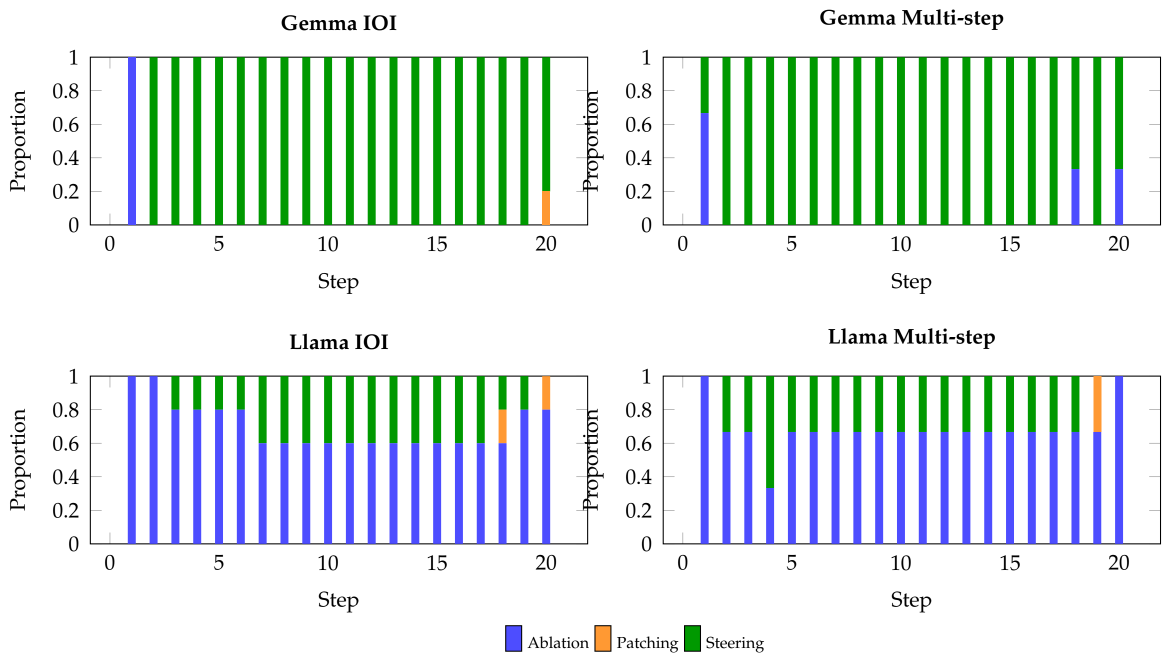

Figure 10: Action type distribution across intervention steps for Gemma-2-2B (top) and Llama-3.2-1B (bottom).

The attribution analysis reveals several structural properties. Gemma-2-2B activates approximately 12 600 transcoder features per IOI prompt, of which roughly 2 200 survive pruning at the 80% influence threshold; Llama-3.2-1B activates approximately 8 000, with roughly 1 600 surviving. On Gemma, causally important features span all 26 layers but concentrate in layers 0–6 and 24–25, whereas on Llama the distribution is more compressed, with early layers (0–4) dominant. By feature count, early-layer features (0–8) form the plurality of the top-10, while the single highest-impact features by KL are consistently found in layers 24–25. Attribution graph generation takes approximately 18 s on Gemma and 12 s on Llama; each `feature_intervention` call takes approximately 0.03 s.

5.5 Multi-step Reasoning

A further test examines whether the agent efficiently identifies features mediating multi-hop reasoning. Three prompts requiring transitive inference or factual chaining are evaluated with the same $B = 20$ budget (Section 4.2.2). Table 5 reports bounded oracle efficiency under the fair ablation-only protocol.

Table 5: Multi-step reasoning under the fair ablation-only protocol: bounded oracle efficiency with bootstrap 95% CIs over 3 prompts.

Model	Method	Mean KL	Median KL	Oracle Eff. [95% CI]
Gemma-2-2B	Oracle	0.000505	—	100.0%
	EAP	0.000495	0.000482	98.1% [97.2, 99.6]
	Greedy	0.000401	0.000455	79.4% [56.6, 94.0]
	Bandit	0.000396	0.000423	78.4% [65.6, 87.4]
	POMDP-abl	0.000369	0.000408	73.0% [63.6, 84.2]
	Random	0.000284	0.000269	56.2% [52.4, 57.4]
Llama-3.2-1B	Oracle	0.012233	—	100.0%
	EAP	0.011938	0.012794	97.6% [62.8, 99.4]
	Bandit	0.009352	0.008046	76.4% [62.5, 89.6]
	Random	0.005476	0.007020	44.8% [39.1, 54.5]
	POMDP-abl	0.001133	0.001003	9.3% [4.4, 55.9]
	+ finer bins	0.004626	—	37.8%
	Greedy	0.001159	0.001074	9.5% [4.7, 56.6]

The oracle's cumulative KL, which forms the efficiency denominator, is 0.0101 on Gemma and 0.2447 on Llama. On Gemma, POMDP-abl (73.0%) again beats random (56.2%) and is competitive with the bandit (78.4%), plain UCB (73.6%), and greedy (79.4%), with EAP strongest (98.1%). On Llama, by contrast, POMDP-abl reaches 9.3%, below random (44.8%) and far below the bandit (76.4%) and plain UCB (75.9%). The per-prompt bounded efficiencies make the failure mode concrete: the agent reaches only 4.4% and 15.2% on the two prompts whose oracle cumulative KL is dominated by early-layer features (oracle CumKL 0.461 and 0.257), and recovers to 55.9% only on the third prompt, whose causal mass sits in later layers (oracle CumKL 0.016). These per-prompt diagnostics trace the collapse to the coarse three-bin layer-role discretisation (Section 4): in Llama's shallow 16-layer stack, the early/middle/late bins each span many functionally distinct layers, so the agent's hidden-state factor cannot localise the early-layer features that mediate multi-hop reasoning. As a targeted remedy, the experiment was re-run with a finer six-bin layer-role discretisation. This roughly quadruples efficiency (9.3% → 37.8%) and confirms the diagnosis, but does not close the gap to the bandit; it is partial evidence that the discretisation, not the EFE objective per se, is the bottleneck, discussed as a limitation in Section 6.

The layer-wise distribution of the top-10 discovered features is contrasted with the IOI case in Figure 11: the multi-step mass shifts visibly toward early and middle layers, whereas the IOI mass remains concentrated in the late layers for both architectures.

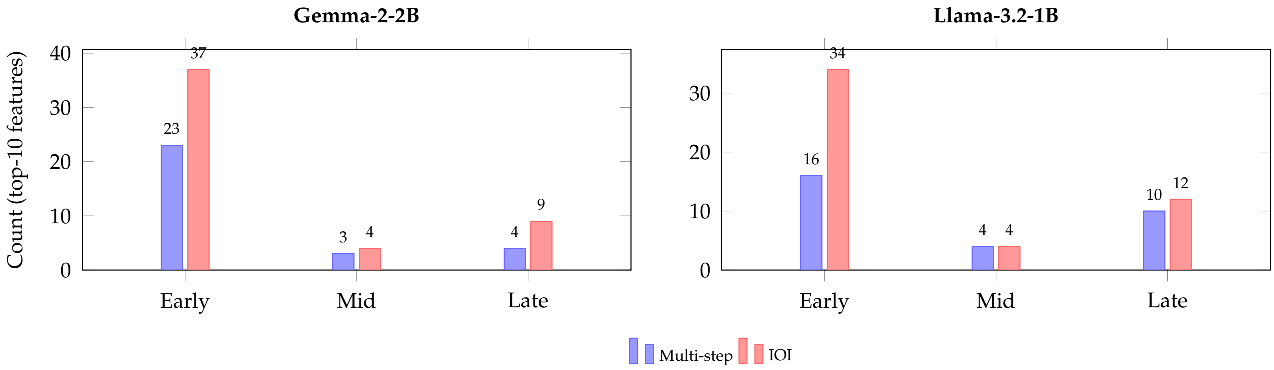

Figure 11: Layer distribution of top-10 causal features for multi-step vs. IOI tasks on both models.

The top causal features for multi-step prompts concentrate in early layers (layers 0–8), consistent with the hypothesis that multi-hop reasoning requires input processing and entity binding in lower layers before final output computation. This contrasts with IOI, where late layers (24–25) dominate.

5.6 Multi-Domain Analysis

Table 6 presents per-domain bounded oracle efficiency across five cognitive categories, each evaluated with two prompts under the same $B = 20$ budget and the fair ablation-only protocol.

Table 6: Multi-domain bounded oracle efficiency (%, ablation-only). EAP and POMDP-abl vs. baselines.

Model	Domain	EAP	POMDP-abl	Bandit	Greedy	Random
Gemma	Geography	99.4	94.0	93.7	96.8	40.9
	Mathematics	99.7	90.7	96.4	18.6	48.6
	Science	89.5	77.6	83.3	56.2	57.4
	Logic	97.5	84.3	54.8	89.1	47.3
	History	98.4	90.4	51.1	71.8	58.8
Llama	Geography	95.2	97.6	93.3	96.5	51.1
	Mathematics	98.3	90.5	87.4	90.6	49.8
	Science	82.4	88.2	67.9	55.5	42.7
	Logic	77.2	66.2	41.3	67.4	53.8
	History	97.9	56.7	55.1	26.4	37.6

Under the fair protocol POMDP-abl is a solid selector across domains on *both* models (77–94% on Gemma, 57–98% on Llama): it beats random in every domain and is competitive with the bandit and greedy, winning on some domains (e.g. Gemma logic/history, Llama science/logic) and trailing on others (e.g. Gemma math/science). EAP remains the strongest single baseline. The domain results thus mirror the IOI picture: the agent reliably beats random and is competitive with greedy/bandit, while EAP attribution ranking is hard to beat. (Absolute multi-action KL per domain is steering-amplified and reported as RCK context, not as oracle efficiency.)

The corresponding per-domain layer distributions are shown in Figure 12, which makes the task-dependent structure explicit: reasoning domains are left-heavy (early layers) while factual domains are right-heavy (late layers) on both models.

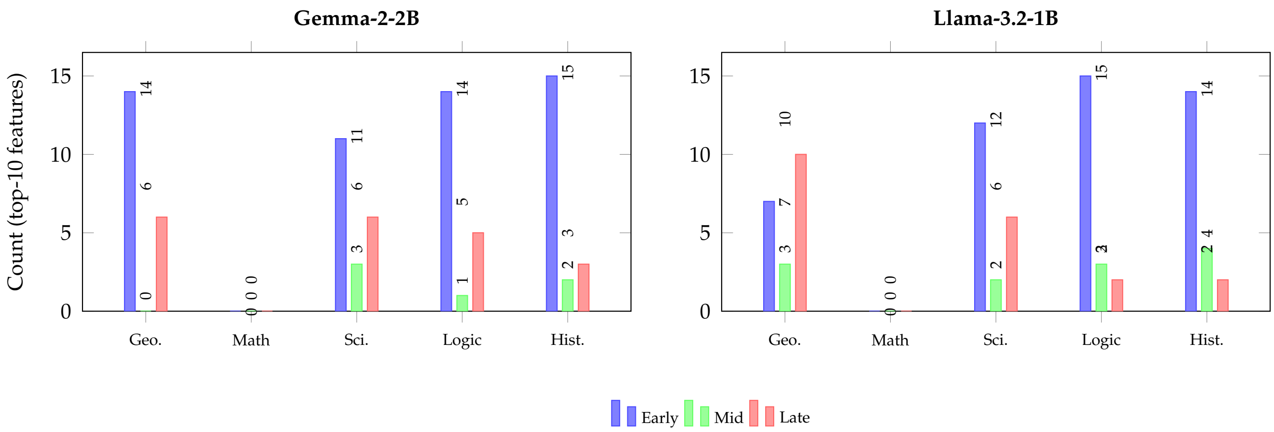

Figure 12: Layer distribution of top-10 causal features across five cognitive domains on both models.

The multi-domain analysis reveals task-dependent circuit structure: reasoning tasks (logic, mathematics) recruit early-layer features, while factual tasks (geography, history) concentrate in late layers. Science prompts show a more uniform distribution.

5.7 Discretisation Sensitivity

The observation model discretises continuous KL and activation magnitudes into bins via fixed thresholds (Section 4). To check that conclusions are not artefacts of these thresholds, Gemma IOI was re-run with the KL and activation thresholds independently scaled by $\pm 20\%$ and POMDP-abl bounded oracle efficiency recomputed (Table 7).

Table 7: Discretisation sensitivity: Gemma IOI POMDP-abl bounded oracle efficiency under $\pm 20\%$ threshold perturbations (baseline 82.0%).

Perturbation	Oracle Eff. (%)
Baseline thresholds	82.0
Activation thresholds $\times 0.8$	86.2
Activation thresholds $\times 1.2$	80.3
KL thresholds $\times 1.2$	82.0
KL thresholds $\times 0.8$	60.2

Efficiency is stable (80–86%) under activation-threshold changes and under a $+20\%$ KL-threshold change, but drops to 60.2% when the KL thresholds are tightened by 20%. The KL discretisation is thus the more sensitive knob: overly fine KL bins inject observation noise into the belief update. The qualitative ordering (POMDP-abl above random, competitive with the bandit, below EAP) is preserved in all five configurations, so the headline conclusions are robust to reasonable threshold choices while the precise efficiency value is not.

A complementary sweep over the agent's three core hyperparameters, the action precision γ, the Dirichlet likelihood learning rate η, and the pragmatic preference weight, confirms that the bounded oracle efficiency is highly stable to these settings (Table 8). On Gemma IOI the efficiency stays at 82.0% across a sixteen-fold range of γ, a four-fold range of η, and a four-fold range of the pragmatic weight. On Llama IOI it is invariant to γ and the pragmatic weight, and varies only mildly with η (49.1–54.5%). The agent's behaviour is therefore governed by the structure of its generative model and the observation stream rather than by fine hyperparameter tuning.

Table 8: Hyperparameter sensitivity of POMDP-abl bounded oracle efficiency (%) on IOI, varying one factor at a time about the default (γ=16, η=1.0, pragmatic weight 1.0).

Factor	Value	Gemma-2-2B	Llama-3.2-1B
Action precision γ	4	82.0	52.1
	64	82.0	52.1
Learning rate η	0.5	82.0	54.5
	2.0	82.0	49.1
Pragmatic weight	0.5	81.9	52.1
	2.0	82.0	52.1

5.8 Cross-Model Validation (Llama-3.2-1B)

To probe generality, all experiments are replicated on Llama-3.2-1B (16 layers, 2048-dim) using transcoders from `mntss/transcoder-Llama-3.2-1B`. Table 9 provides a consolidated head-to-head of the four ablation-only selectors against the EAP and ablation oracles across all three task families and both architectures.

Table 9: Head-to-head bounded oracle efficiency (%, ablation-only) across IOI, multi-step, and domain tasks for both models. Domain values are means over the five cognitive domains.

Task	Method	Gemma-2-2B	Llama-3.2-1B
IOI	EAP	91.9	95.2
	POMDP-abl	82.0	52.1
	Bandit	74.4	88.2
	UCB	73.7	55.2
Multi-step	EAP	98.1	97.6
	POMDP-abl	73.0	9.3
	Bandit	78.4	76.4
	UCB	73.6	75.9
Domain	EAP	96.9	90.2
	POMDP-abl	87.4	79.9
	Bandit	75.9	69.0
	UCB	77.1	66.7

The agent transfers cleanly to Gemma's deeper stack, where POMDP-abl beats random, edges out both the heuristic bandit and plain UCB, and is the strongest non-EAP selector across IOI and the domain tasks. On Llama it transfers only partially: it remains competitive on the single-step domain tasks (79.9%, above both bandit and UCB), but on IOI its efficiency (52.1%) is only marginally above random (48.4%) and well below the bandit (88.2%) and EAP (95.2%), and the coarse layer-role discretisation hurts it badly on multi-step reasoning (Section 5.5). The heuristic bandit and the plain UCB baseline track each other closely on Gemma (both near 74–78%), indicating that the bandit's hand-built importance and locality priors add little there. On Llama IOI, by contrast, the heuristic bandit (88.2%) far exceeds plain UCB (55.2%), showing that those priors carry the bandit on the shallower architecture. Overall the agent is competitive with, and often slightly ahead of, the action-matched baselines on Gemma, but behind the heuristic bandit on the shallower Llama stack.

5.9 Hypothesis Evaluation

The four pre-registered hypotheses (H1–H4) are evaluated against the experimental evidence and summarised in Table 10. H2 is split into manipulability (H2a) and selectivity (H2b) following the steering controls above.

Table 10: Hypothesis evaluation summary. H1/H3 shown as Gemma / Llama.

Hypothesis	Criterion	Result	p-value	Outcome
H1: Efficiency	POMDP-abl > Random (paired perm.)	+43.5% / +7.8%	0.031 / 0.41	Accepted (Gemma); not on Llama
H2a: Manipulab.	Steering changes > chance	11/50 & 9/50 at m=10	$< 10^{-8}$	Accepted
H2b: Selectivity	Selected > control (Fisher)	11 vs 8; 9 vs 6	n.s.	Not supported
H3: Discovery	Bounded oracle eff. ≥ 50%	82.0% / 52.1% (IOI)	—	Accepted (IOI); fails Llama multi-step
H4: Transfer	Qualitative replication	Partial (Gemma full, Llama partial)	—	Partially accepted

Hypothesis H1 (efficiency) is accepted on Gemma but not on Llama. Under the fair ablation-only protocol, POMDP-abl beats random on Gemma IOI (+43.5%, paired permutation $p = 0.031$, $n = 5$) but the gain on Llama IOI is not significant (+7.8%, $p = 0.41$). No claim is made of beating EAP, which is the strongest ablation baseline throughout.

Hypothesis H2 (causal controllability) is partially supported. H2a (manipulability) holds, as steering produces dose-dependent prediction changes far above chance ($p < 10^{-8}$ at m=10). H2b (selectivity) is not supported: circuit-selected features are only marginally, and not significantly (Fisher exact, n.s.), more likely to change predictions than random active features, and the largest multipliers are off-distribution.

Hypothesis H3 (discovery quality) is accepted except for Llama multi-step. Bounded oracle efficiency exceeds the 50% criterion on Gemma IOI (82.0%), Gemma multi-step (73.0%), Llama IOI (52.1%) and all domain tasks, and falls to 9.3% on Llama multi-step (37.8% with finer bins).

Hypothesis H4 (cross-architecture transfer) is partially accepted. Findings replicate fully on Gemma and partially on Llama, where the agent is competitive on IOI and domain tasks but the layer-role discretisation reduces multi-step performance.

Limitations of the present work are discussed in Section 6.

5.10 Comparison with Published Circuit Discovery Methods

Table 11 positions ACD against published circuit discovery methods on the IOI benchmark. ACD is not a competing graph-construction method: it consumes an attribution graph (here from `circuit-tracer`/EAP) and adds an adaptive intervention-selection policy on top. Prior methods report faithfulness or node-ablation accuracy rather than oracle efficiency, so the table summarises the closest analogous metric and evaluation model and should be read as context, not a like-for-like leaderboard. ACD's bounded oracle efficiency (ablation-only, $\leq 100\%$) is reported as the comparable quantity, with the multi-action RCK noted separately.

Table 11: Contextual comparison of ACD with published circuit discovery methods on IOI.

Method	Approach	Model	Budget B	Adapt.	Oracle Eff.
ACDC [8]	Exhaustive edge pruning	GPT-2 Small: 117M	$O(E)$, ∼5k–50k	Thresh.	N/A
EAP [9]	Gradient attribution	GPT-2 Small: 117M	$O(E)$, single pass	No	N/A
Causal Tracing [10]	Activation patch sweep	GPT (up to 175B)	$O(L \times T)$	No	N/A
Sparse Feature Circuits [11]	EAP + SAE features	Pythia 2.8B	$O(E)$, single pass	No	N/A
Dict. Learning Circuits [12]	Patch-free + dict. learning	Various ≤7B	$O(E)$, single pass	No	N/A
EAP ranking (this backend)	Attribution-ranked ablation	Gemma-2-2B; Llama-3.2-1B	$B = 20$	No	91.9%; 95.2%
ACD POMDP-abl (this work)	POMDP-EFE, ablation only	Gemma-2-2B; Llama-3.2-1B	$B = 20$	Yes	82.0%; 52.1%

Within this backend, a direct EAP attribution ranking (91.9%/95.2%) is the strongest ablation-KL selector and is not beaten by the EFE agent; POMDP-abl (82.0%/52.1%) reliably beats random and is competitive with greedy, the heuristic bandit, and a plain UCB baseline. The agent's distinctive value is therefore not raw efficiency but its adaptive, uncertainty-aware policy: explicit belief-state uncertainty, a multi-action intervention space, and online generative-model learning, none of which the static baselines provide. The multi-action RCK (1255% on Gemma IOI) reflects steering-driven amplification, analysed in Section 5.2, and is not a discovery-efficiency claim. Prior methods are not directly comparable because they optimise faithfulness rather than oracle-KL efficiency under a fixed budget.

6 Discussion

6.1 Action Selection Dynamics

The temporal pattern of action selection reveals the exploration–exploitation trade-off predicted by Expected Free Energy theory. On Gemma IOI, the agent selects ablation at the first step for its maximal epistemic value, then transitions predominantly to feature steering, which carries lower epistemic but high pragmatic value (exact per-step tallies in Figure 10). This progression is grounded in the action-conditioned B-matrix (Appendix B): ablation produces the highest transition entropy, while steering produces the lowest. Activation patching appears only sporadically at intermediate confidence levels. Notably, this distribution is not prescribed but emerges from the generative model, suggesting that the EFE objective provides a principled mechanism for adaptive experiment design.

6.2 Structure Discovery vs. Steering-KL Exploitation

A fair reading of the action dynamics above raises the question of whether the agent's headline performance reflects circuit-structure discovery or the exploitation of steering, which mechanically produces larger KL shifts than ablation. Two design choices address this directly. First, the primary discovery-quality metric is the ablation-only agent (POMDP-abl) scored by bounded oracle efficiency against the ablation oracle (Equation 14); here the agent and the oracle use the identical intervention, so any advantage is attributable to selection, not to the steering action. Second, the action-matched baselines (Greedy+steer, Bandit+steer, Random+steer) grant the heuristics the same steering action and are reported on the common RCK scale (Equation 15). The decomposition in Section 5 shows that a substantial part of the full multi-action agent's RCK above 100% is contributed by the steering action itself, since random selection with steering already amplifies cumulative KL relative to the ablation oracle; that amplification is therefore not presented as evidence of better structure discovery. The evidence that EFE-guided selection helps comes from the bounded POMDP-abl-versus-oracle comparison on the same ablation footing, where on Gemma POMDP-abl significantly beats random and is competitive with greedy and the bandit, rather than from the inflated RCK numbers. This selection advantage is model-dependent: on Llama, POMDP-abl only matches random and trails the bandit and EAP, so the case for EFE selection rests primarily on the Gemma results.

6.3 Theoretical Connections

The EFE objective [17] provides a principled probabilistic foundation distinct from heuristic baselines. Its epistemic term measures information gain (mutual information between hidden states and observations), while its pragmatic term measures goal satisfaction (preference for high causal importance). This decomposition aligns naturally with circuit discovery, where features that are both causally important and whose role is uncertain should be prioritised. As noted by Tschantz et al. [38], EFE minimisation reduces to reward maximisation or pure information gain in limiting cases; the present experiments operate in a regime where both terms contribute meaningfully.

The EFE objective also bears a deep connection to classical Bayesian optimal experimental design [13]. The ACD framework extends this classical formulation through a multi-factor hidden state model capturing transformer-specific structure, the incorporation of pragmatic value alongside epistemic value, and online Dirichlet concentration parameter learning that enables the agent to refine its observation model during discovery.

6.4 Task-Dependent Circuit Structure

A striking finding is the strong task-dependence of circuit layer distribution. Factual tasks (IOI, geography, history) concentrate in late layers, while reasoning tasks (multi-step, logic, mathematics) recruit early-layer features. The POMDP agent's explicit layer-role state factor naturally adapts to these patterns through belief updating, providing an adaptive structure-learning capacity absent in fixed-priority heuristics.

6.5 Cross-Model Generality

Replication on Llama-3.2-1B [40] indicates that the framework transfers only partially across architecture families. The agent remains a useful selector on the Llama domain benchmark (competitive with the bandit and EAP), but on Llama IOI its bounded oracle efficiency is modest (52.1%, CI [17.4, 90.2]), barely above random (48.4%) and clearly below the bandit (88.2%) and EAP (95.2%). The word "dominates" is deliberately avoided: on several Llama conditions the bandit tracks or beats the agent (Fig. 4), and these are reported as ties or losses rather than wins. The one task where the agent underperforms random, Llama multi-step, is examined as a limitation in its own right below.

6.6 Limitations

The primary limitation concerns multi-step reasoning on Llama. On Llama multi-step the agent underperforms random selection (Table 5); this outcome is reported as a limitation rather than attributed to the architecture. Per-prompt diagnostics (Section 5) show the failure concentrates on the prompts whose high-KL features sit in adjacent middle layers that the default three-bin layer-role factor collapses into a single "middle" state, so the agent cannot distinguish them. As a concrete remedy, the benchmark was re-run with the layer-role factor refined to six bins (`--n-layer-role 6`); the outcome is reported in Section 5. Even with the remedy, multi-hop reasoning on shallow architectures remains the weakest setting for the current generative model, and architecture-specific or learned layer groupings are needed before the method can be relied upon there.

A further limitation is the small prompt set and its effect on significance. The evaluation uses 5 IOI, 3 multi-step, and 10 domain prompts. Standard practice favours $n \geq 30$ for reliable significance testing; to compensate, bootstrap 95% confidence intervals and an exact paired-permutation test are reported alongside the t-test (Section 4.6). Where the permutation test is not significant (for example, Llama IOI) H1 is not accepted. Expanding to 30 or more prompts per condition remains important future work.

A final limitation concerns discretisation and hyperparameter sensitivity. Observation thresholds were calibrated on Gemma. A $\pm 20\%$ threshold sweep on Gemma IOI (Table 7) shows bounded efficiency stable (80–86%) under activation-threshold and $+20\%$ KL-threshold changes, while tightening the KL thresholds by 20% drops it to 60.2%, identifying the KL discretisation as the sensitive knob. By contrast, a sweep over the agent's core hyperparameters (γ, η, pragmatic weight; Table 8) leaves the efficiency essentially unchanged, so the conclusions are not an artefact of hyperparameter tuning. The qualitative ordering (above random, competitive with the bandit, below EAP) is preserved in all configurations, but adaptive quantile-based or learned discretisation boundaries would improve robustness and portability.

A second limitation concerns comparison scope and the absence of a direct ACDC run. The strongest established comparison reported here is a direct EAP attribution ranking computed on the same `circuit-tracer` backend, which is consistently competitive with or stronger than the agent. A direct ACDC run is not included because ACDC is defined over the attention-head and MLP edge graph of a model and does not operate on the transcoder-feature candidate set or the fixed per-feature intervention budget used here, so a like-for-like ACDC comparison would require a different evaluation substrate. This is a real limitation of the present comparison rather than evidence of superiority, and porting the agent to an edge-level graph where ACDC applies is left to future work.

The intervention protocol also operates at a single token position. Interventions and the KL metric are evaluated at the final next-token position, so features whose causal role is specific to earlier token positions are not separately localised; a position-resolved observation model is a natural extension. A related caveat is selection bias in the candidate set, since candidates are drawn from the top features by graph importance. The evaluation therefore measures selection efficiency within an already-pruned set rather than over all model components, and a different pruning threshold could shift the absolute efficiencies.

Two further limitations concern the agent itself. Single-step planning (policy length one) prevents reasoning about complementary intervention sequences; multi-step planning via tree search with pruning could capture synergistic effects at the cost of combinatorial complexity. Finally, the preference model assumes that high KL necessarily indicates causal importance, which may not hold for tasks with spurious correlations, so adaptive or learned preference models could improve generalisation.

6.7 Implications for Safety Auditing

The ACD framework has direct implications for AI safety auditing [41]. The active inference architecture automatically directs intervention budget toward maximally informative features, the multi-action capability provides multiple perspectives on causal influence, and the online learning mechanism enables adaptation to new domains. In practice, safety engineers could deploy the POMDP agent to identify causally necessary features within a fixed budget, enabling verification that reasoning circuits match expected pathways and detection of shortcut dependencies. The framework also connects to knowledge editing [10]: by identifying mediating features, targeted circuit modifications become feasible without full model retraining.

6.8 Broader Implications

The ACD framework demonstrates that active inference, a theory of rational agency from computational neuroscience, provides both theoretical motivation and practical benefits for circuit discovery [1, 41]. Circuit discovery is formulated as adaptive Bayesian experimental design [13], where the agent's exploration–exploitation behaviour emerges from the Free Energy Principle rather than being hand-tuned. The framework scales from toy tasks (GPT-2 Small) to production-size models (2B+ parameters). On Gemma, the full multi-action agent attains a Relative Cumulative KL above 100% (Equation 15); this reflects the steering action amplifying measurable causal impact relative to the ablation oracle, not a super-oracle "efficiency", and the discovery-quality claims are grounded in the bounded POMDP-abl comparison instead. Future work should investigate scaling to larger models with task-adaptive priors, integrating additional modalities, and applying active inference to other interpretability tasks such as deception detection.

7 Conclusions and Future Work

This paper presented Active Circuit Discovery (ACD), a framework that combines attribution-graph analysis with active inference for efficient intervention selection in large language models. The core contribution is the formulation of circuit discovery as a POMDP with a three-action intervention space, solved by a multi-factor Active Inference agent implemented with pymdp [18]. The agent maintains beliefs over three hidden-state factors (feature importance, layer role, and causal influence), selects both the target feature and intervention type (ablation, activation patching, or feature steering) by minimising Expected Free Energy over the joint feature–action space, and learns its observation model online through Dirichlet parameter updates from real intervention data. The framework integrates Anthropic's `circuit-tracer` library [19] for Edge Attribution Patching with transcoders, and all interventions use the `feature_intervention` API at the transcoder level.

The evaluation spanned two architectures (Gemma-2-2B [35] and Llama-3.2-1B [40]) and four benchmarks: IOI, multi-step reasoning, feature steering, and a multi-domain benchmark spanning geography, mathematics, science, logic, and history. The agent was compared against a heuristic UCB-style bandit, a plain UCB baseline, greedy importance ranking, a direct EAP attribution ranking, random selection, action-matched (steering-enabled) variants of these heuristics, and an oracle upper bound, with bootstrap 95% confidence intervals reported throughout.

The four hypotheses stated in Section 1 are resolved as follows. Hypothesis H1 (efficiency) is accepted on Gemma, where the ablation-only agent exceeds random selection by 43.5 percentage points on IOI under both a paired t-test and an exact permutation test ($p = 0.031$); on Llama IOI the effect is not significant, so H1 is not accepted there. Hypothesis H2 (causal controllability) is partially supported: scaling features changes the top-1 prediction far above chance and dose-dependently ($p < 10^{-8}$ at the largest multiplier, H2a accepted), but circuit-selected features are not significantly more steerable than matched random features (H2b not accepted), so part of the steering effect is attributable to generic activation scaling. Hypothesis H3 (discovery quality), assessed by the bounded oracle efficiency of the ablation-only agent against the ablation oracle, is accepted on Gemma IOI (82.0%), Gemma multi-step (73.0%), Llama IOI (52.1%), and all domain tasks, and reaches 9.3% on Llama multi-step (37.8% with finer layer-role bins). Hypothesis H4 (cross-architecture transfer) holds qualitatively, with the caveat that the bandit ties the agent on several Llama conditions and that Llama multi-step is the weakest setting (Section 6).

A consistent finding across both architectures is the task-dependent layer structure of circuits: IOI circuits concentrate in late layers, reasoning and logic features concentrate in early layers, and science prompts show a more uniform distribution. The full multi-action agent attains a Relative Cumulative KL above 100% on Gemma, reported as a steering-driven amplification factor (Equation 15) distinct from the bounded efficiency used for the discovery-quality claims.

Future work will increase prompt-set sizes for statistical power, explore multi-step planning in which the agent composes sequences of complementary interventions, investigate adaptive discretisation of the observation space, and add a direct ACDC head-to-head once a shared candidate set is available. The open-source codebase, Colab notebooks (free T4 GPU), and both aarch64 (DGX Spark) and standard `x86_64/amd64` Docker images are released to facilitate reproduction.

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Abbreviations:

The following abbreviations are used in this manuscript:

ACD	Active Circuit Discovery
ACDC	Automated Circuit Discovery
EAP	Edge Attribution Patching
EFE	Expected Free Energy
FEP	Free Energy Principle
IOI	Indirect Object Identification
KL	Kullback–Leibler (divergence)
LLM	Large Language Model
MLP	Multilayer Perceptron
POMDP	Partially Observable Markov Decision Process
SAE	Sparse Autoencoder

Appendix

A Active Inference Selector Parameters

This appendix details the hyperparameters of the Active Inference selector and the heuristic bandit baseline.

The POMDP agent is instantiated with the pymdp parameters listed in Table A1. Action selection uses a Boltzmann policy with precision $\gamma = 16.0$ over the negative Expected Free Energy. The exploration behaviour of the agent is not controlled by a single scalar but emerges from the two information-gain terms of the Expected Free Energy, namely state information gain and Dirichlet parameter information gain, both of which are enabled. The observation likelihood is learned online with a Dirichlet learning rate $\eta = 1.0$, and the pragmatic preference weight on the preference vector is 1.0. The hidden state comprises four

Table A1: POMDP agent hyperparameters (pymdp).

Symbol	Meaning	Value
γ	action precision (policy temperature)	16.0
η	Dirichlet likelihood learning rate (lr_{pA})	1.0
w_{prag}	pragmatic preference weight on $\boldsymbol{C}$	1.0
state info gain	epistemic term over hidden states	enabled
param info gain	epistemic term over Dirichlet parameters	enabled
policy length	planning horizon	1
$\lvert\mathcal{S}\rvert$	hidden-state factors (importance, role, causal)	$4 \times 3 \times 3$
κ	importance-to-KL prior scaling	0.01

importance levels, three layer-role bins, and three causal-strength levels; the policy horizon is one step.

Before an intervention is executed, the agent derives a prior observation for each candidate from its graph metadata. The normalised graph importance $\mathrm{imp}(i) \in [0, 1]$ is scaled by a factor of $\kappa = 0.01$ before discretisation: $o_{\mathrm{prior}} = \mathrm{discretise}(\mathrm{imp}(i) \times \kappa)$. This maps the graph-derived importance into the KL divergence threshold range $[10^{-4}, 10^{-2}]$, so that the prior observation reflects expected KL magnitudes rather than raw graph weights.

Candidates are extracted as up to 5 features per layer, 60 total, selected from pruned graph features sorted by influence (sum of absolute adjacency weights). The pruning thresholds are node $= 0.8$ and edge $= 0.98$, the defaults from `circuit-tracer`.

The heuristic bandit baseline uses a separate set of parameters that are not part of the active-inference agent. Its exploration weight is $\omega_e = 2.0$, applied to a per-feature uncertainty bonus. All features start with uncertainty $u(i) = 1$; after feature i is observed its uncertainty drops to 0, same-layer features receive a 30% reduction ($u(j) \leftarrow 0.7\, u(j)$), and adjacent-position features receive a 10% reduction ($u(j) \leftarrow 0.9\, u(j)$). A layer prior is initialised to $\lambda_\ell = 1$ for all layers and updated after each observation as $\lambda_\ell = 1 + 0.5(\bar{\mathrm{KL}}_\ell / \bar{\mathrm{KL}}_{\mathrm{global}} - 1)$. The plain UCB baseline removes these engineered priors entirely and applies the textbook UCB1 rule over layers using the same exploration weight.

B B-Matrix Transition Priors

The transition model $\mathbf{B}$ for Factor 0 (feature importance) uses action-conditioned dynamics that encode the causal semantics of each intervention type. These are calibrated priors informed by the distinct informational roles of ablation, activation patching, and feature steering.

Ablation (action 0) is an exploratory intervention that removes a feature entirely, producing the broadest distribution of possible outcomes. Its transition matrix has the lowest diagonal concentration (50% stay, 25% down, 25% up), yielding the highest transition entropy among the three actions. Under the EFE decomposition, this high entropy translates into high expected information gain, making ablation the preferred action when the agent is uncertain about a feature's importance.

Activation patching (action 1) replaces the feature activation with a reference value, providing a moderately informative signal. The transition is more concentrated (70% stay, 15% down, 15% up), producing moderate information gain. Patching is selected when the agent has partial evidence and seeks to refine its importance estimate.

Feature steering (action 2) scales the feature activation by a multiplier while preserving its directional contribution. This near-identity transition (90% stay, 5% down, 5% up) has the lowest transition entropy, making it the preferred action when the agent's belief is already concentrated and confirmation is sought rather than exploration.

All transition probabilities are symmetric around the current state to ensure that the expected utility is similar across actions, so that the epistemic term (information gain) drives action selection.

The three diagonals are not tuned to maximise a metric; they encode a deliberate ordering of transition entropy that mirrors the operational semantics of the interventions. Ablation deletes a feature and is the most disruptive, so it should admit the widest range of importance revisions (lowest diagonal, highest entropy ⇒ largest epistemic value); patching substitutes a reference activation and is intermediate; steering merely rescales an existing direction and is the gentlest, so it should mostly preserve the current importance estimate (highest diagonal, lowest entropy). What matters for action selection is this monotone entropy ordering $H(\mathbf{B}_{\text{abl}}) > H(\mathbf{B}_{\text{patch}}) > H(\mathbf{B}_{\text{steer}})$, not the exact fractions. The specific values $(0.50, 0.70, 0.90)$ are the simplest round numbers that realise this ordering while keeping all rows valid tridiagonal distributions; the boundary states absorb the missing mass, giving the corner diagonals of 0.75/0.85/0.95 shown in Appendix C.

The 0.01 importance-scaling factor in Appendix A is not arbitrary: normalised graph importance lies in $[0, 1]$, whereas observed ablation KL on these models lies in roughly $[10^{-4}, 10^{-2}]$. Multiplying by 0.01 maps the top of the importance range onto the top of the empirical KL range so that the graph-derived *prior* observation is discretised into the same bins the agent will later see from real interventions; without it, every candidate would saturate the "large-KL" bin and the prior would be uninformative.

Because these thresholds are fixed rather than learned, their influence is tested directly. Section 4.6 re-runs Gemma IOI with the KL and activation discretisation thresholds scaled by $\pm 20\%$; the resulting bounded oracle efficiencies are reported in Table 7. Efficiency is stable (80–86%) under activation-threshold changes and a $+20\%$ KL-threshold change, but tightening the KL thresholds by 20% lowers it to 60.2%, so the KL discretisation is the sensitive knob; crucially, the qualitative ordering (above random, competitive with the bandit, below EAP) is preserved in all configurations. A full joint hyperparameter sweep is left to future work.

C Complete Generative Model Specification

For full reproducibility this section lists the exact generative model used by the default agent (26-layer Gemma configuration), auto-generated from the implementation by `scripts/dump_generative_model.py`. The hidden state comprises 4 importance levels × 3 layer roles × 3 causal-influence levels; observations comprise 4 KL levels, 4 activation levels, and 3 connectivity levels. The action-conditioned importance transition tensor $\mathbf{B}^{(0)}$, the identity transitions $\mathbf{B}^{(1)}$ and $\mathbf{B}^{(2)}$ for the intrinsic factors, the preference vectors $\mathbf{C}$, the state priors $\mathbf{D}$, all three observation likelihood modalities $\mathbf{A}^{(0)}, \mathbf{A}^{(1)}, \mathbf{A}^{(2)}$ (each tabulated over its non-trivial conditioning factors), and the Dirichlet concentration prior are given below. The complete numeric arrays are also released as `results/generative_model.json`.

The transition model $\mathbf{B}^{(0)}$ for the importance factor is action-conditioned. Columns are the current importance state, rows the next state, for each intervention action:

$$\mathbf{B}^{(0)}_{\text{ablation}} = \begin{bmatrix} 0.750 & 0.250 & 0.000 & 0.000 \\ 0.250 & 0.500 & 0.250 & 0.000 \\ 0.000 & 0.250 & 0.500 & 0.250 \\ 0.000 & 0.000 & 0.250 & 0.750 \end{bmatrix}$$

$$\mathbf{B}^{(0)}_{\text{activation patching}} = \begin{bmatrix} 0.850 & 0.150 & 0.000 & 0.000 \\ 0.150 & 0.700 & 0.150 & 0.000 \\ 0.000 & 0.150 & 0.700 & 0.150 \\ 0.000 & 0.000 & 0.150 & 0.850 \end{bmatrix}$$

$$\mathbf{B}^{(0)}_{\text{feature steering}} = \begin{bmatrix} 0.950 & 0.050 & 0.000 & 0.000 \\ 0.050 & 0.900 & 0.050 & 0.000 \\ 0.000 & 0.050 & 0.900 & 0.050 \\ 0.000 & 0.000 & 0.050 & 0.950 \end{bmatrix}$$

The layer-role factor $\mathbf{B}^{(1)}$ and the causal-strength factor $\mathbf{B}^{(2)}$ are intrinsic properties with identity (single-action) dynamics:

$$\mathbf{B}^{(1)} = \begin{bmatrix} 1.000 & 0.000 & 0.000 \\ 0.000 & 1.000 & 0.000 \\ 0.000 & 0.000 & 1.000 \end{bmatrix}, \qquad \mathbf{B}^{(2)} = \begin{bmatrix} 1.000 & 0.000 & 0.000 \\ 0.000 & 1.000 & 0.000 \\ 0.000 & 0.000 & 1.000 \end{bmatrix}.$$

The preference vectors $\mathbf{C}$ encode log-preferences over observations (higher KL and activation are more informative):

$$\mathbf{C}^{(\text{KL})} = \begin{bmatrix} 0.000 & 1.000 & 3.000 & 5.000 \end{bmatrix}, \qquad \mathbf{C}^{(\text{act})} = \begin{bmatrix} 0.000 & 0.500 & 2.000 & 4.000 \end{bmatrix},$$
$$\mathbf{C}^{(\text{conn})} = \begin{bmatrix} 0.000 & 1.000 & 2.000 \end{bmatrix}.$$

The priors $\mathbf{D}$ over the three hidden-state factors are:

$$\mathbf{D}^{(\text{imp})} = \begin{bmatrix} 0.400 & 0.300 & 0.200 & 0.100 \end{bmatrix}, \qquad \mathbf{D}^{(\text{role})} = \begin{bmatrix} 0.333 & 0.333 & 0.333 \end{bmatrix},$$
$$\mathbf{D}^{(\text{causal})} = \begin{bmatrix} 0.500 & 0.300 & 0.200 \end{bmatrix}.$$

The observation likelihood $\mathbf{A}^{(0)}$ for the KL modality is independent of layer role; each row gives $P(o^{\text{KL}} \mid s_{\text{imp}}, s_{\text{causal}})$ over the four KL levels (negligible, small, medium, large):

$\mathbf{A}^{(0)} =$

imp.	causal	neg.	small	med.	large
negligible	weak	0.631	0.243	0.078	0.049
negligible	moderate	0.512	0.237	0.143	0.107
negligible	strong	0.375	0.225	0.205	0.195
low	weak	0.558	0.242	0.122	0.078
low	moderate	0.421	0.229	0.184	0.166
low	strong	0.283	0.217	0.247	0.253
moderate	weak	0.467	0.233	0.163	0.137
moderate	moderate	0.329	0.221	0.226	0.224
moderate	strong	0.192	0.208	0.288	0.312
high	weak	0.375	0.225	0.205	0.195
high	moderate	0.237	0.212	0.268	0.282
high	strong	0.100	0.200	0.330	0.370

The activation modality $\mathbf{A}^{(1)}$ depends on importance and layer role (it is independent of causal strength); each row gives $P(o^{\text{act}} \mid s_{\text{imp}}, s_{\text{role}})$ over the four activation levels (inactive, low, moderate, high):

$\mathbf{A}^{(1)} =$

imp.	role	inact.	low	mod.	high
negligible	early	0.550	0.250	0.120	0.080
negligible	middle	0.550	0.250	0.120	0.080
negligible	late	0.505	0.250	0.140	0.105
low	early	0.400	0.250	0.187	0.163
low	middle	0.400	0.250	0.187	0.163
low	late	0.355	0.250	0.207	0.188
moderate	early	0.250	0.250	0.253	0.247
moderate	middle	0.250	0.250	0.253	0.247
moderate	late	0.205	0.250	0.273	0.272
high	early	0.100	0.250	0.320	0.330
high	middle	0.100	0.250	0.320	0.330
high	late	0.055	0.250	0.340	0.355

The connectivity modality $\mathbf{A}^{(2)}$ depends on layer role and causal strength (it is independent of importance); each row gives $P(o^{\text{conn}} \mid s_{\text{role}}, s_{\text{causal}})$ over the three connectivity levels (sparse, moderate, dense):

$\mathbf{A}^{(2)} =$

role	causal	sparse	mod.	dense
early	weak	0.560	0.300	0.140
early	moderate	0.460	0.300	0.240
early	strong	0.360	0.300	0.340
middle	weak	0.500	0.300	0.200
middle	moderate	0.400	0.300	0.300
middle	strong	0.300	0.300	0.400
late	weak	0.560	0.300	0.140
late	moderate	0.460	0.300	0.240
late	strong	0.360	0.300	0.340

The observation model is learned online. The Dirichlet concentration parameters are initialised as $\mathbf{a}^{(m)} = 10\,\mathbf{A}^{(m)}$ for each modality m (a concentration of ten times the prior likelihood), and updated with learning rate $\eta = 1.0$ after every intervention.

D Correspondence Metric

The primary metric for evaluating selector quality is the *bounded oracle efficiency* (Equation 14): the ratio of cumulative *ablation* KL achieved by an ablation-only selector to that achieved by the ablation oracle (features sorted by true ablation KL, descending):

$$\eta = \frac{\sum_{t=1}^{B} \mathrm{KL}^{\mathrm{abl}}_{i_t^{\mathrm{method}}}}{\sum_{t=1}^{B} \mathrm{KL}^{\mathrm{abl}}_{i_t^{\mathrm{oracle}}}} \times 100\% \in [0, 100]\%. \tag{D1}$$

Both numerator and denominator are ablation KL, so η is bounded by 100%. For the full multi-action agent the Relative Cumulative KL is reported instead (RCK, Equation 15), which divides the agent's (possibly steering-inflated) cumulative KL by the same ablation-oracle denominator and may therefore exceed 100%; it is an amplification factor, never an efficiency.

Secondary metrics include mean KL per intervention (higher = better feature selection), its median and inter-quartile range (reported because the per-prompt coefficient of variation is high), and improvement percentages over random and greedy baselines.

KL divergence between clean and intervened output distributions is computed as:[1]

$$D_{\mathrm{KL}}(p_{\mathrm{clean}} \| p_{\mathrm{interv}}) = \sum_{v} p_{\mathrm{clean}}(v) \log \frac{p_{\mathrm{clean}}(v)}{p_{\mathrm{interv}}(v)} \tag{D2}$$

using `torch.nn.functional.kl_div` with a 10^{-10} smoothing factor to prevent numerical instability.

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